

Lecture Note in Calculus B - 2nd

Dr. Fares Al-Azemi

Done by: Haneen Alfadhli

@HAlfadhli17

11.1-Infinite Sequences

Def. [infinite sequence]:

An infinite sequence of numbers is a function whose domain is the set of positive integers [Natural number]; it is an expression in the form:

$$\boxed{a_1}, a_2, a_3, \dots, a_{n-1}, \boxed{a_n}, a_{n+1}, \dots = \{a_n\}_{n=1}^{\infty}$$

First term

nth term

Examples

1. $2, 4, 6, 8, \dots, 2n, \dots = \{2n\}_{n=1}^{\infty}$

Sequence of even number, increasing sequence.

2. $\left\{\frac{1}{n}\right\}_{n=1}^{\infty}$

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{n}, \dots$$

Decreasing sequence, goes to zero.

3. $1, 3, 5, 7, 9, \dots, 2n-1, \dots = \{2n-1\}_{n=1}^{\infty}$

Sequence of odd numbers, increasing sequence.

4. $\left\{\frac{n-1}{n}\right\}_{n=1}^{\infty}$

$$= 0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots, \frac{n-1}{n}, \dots$$

Increasing sequence, goes to one.

$$5. \{5\}_n^\infty = 1$$

$$= 5, 5, 5, 5, \dots, 5, \dots$$

Constant sequence, goes to 5.

$$6. \left\{(-1)^{n+1} \frac{1}{n}\right\}_n^\infty = 1$$

$$= 1, \frac{-1}{2}, \frac{1}{3}, \frac{-1}{4}, \frac{1}{5}, \dots, (-1)^{n+1} \frac{1}{n}, \dots$$

Neither increasing nor decreasing sequence.

$$7. \text{The sequence with } a_1 = 1, a_{n+1} = \frac{a_n}{n+1}$$

$$a_1 = 1$$

$$, a_2 = \frac{a_1}{1+1} = \frac{1}{2}$$

$$, a_3 = \frac{\frac{1}{2}}{2+1} = \frac{1}{6}$$

$$, a_4 = \frac{\frac{1}{6}}{3+1} = \frac{1}{24}, \dots, a_n = \frac{1}{n!} = \left\{\frac{1}{n!}\right\}_n^\infty = 1$$

Decreasing sequence, goes to zero.

Def. [n factorial]:

The factorial of a non-negative integer n, denoted by n! , is the product of all +ve integers less than or equal to n.

$$n! = n \times (n-1) \times (n-2) \times (n-3) \dots 3 \times 2 \times 1.$$

Note that: $0! = 1$, $\infty! = \infty$,

$$n! < n^n, \forall n > 0$$

Def. [convergence of sequences]:

If $\lim_{n \rightarrow \infty} a_n = L$ (exist) we say that $\{a_n\}$ converge.

If $\lim_{n \rightarrow \infty} a_n =$ fails to exist ($\begin{smallmatrix} DNE \\ \pm \infty \end{smallmatrix}$) we say that $\{a_n\}$ diverge.

Back to previous *Examples*

1. $\lim_{n \rightarrow \infty} 2n = \infty \rightarrow \{2n\} \text{ is div.}$

2. $\lim_{n \rightarrow \infty} \frac{1}{n} = 0 \rightarrow \left\{\frac{1}{n}\right\} \text{ is conv.}$

3. $\lim_{n \rightarrow \infty} (2n - 1) = \infty \rightarrow \{2n - 1\} \text{ is div.}$

4. $\lim_{n \rightarrow \infty} \left(\frac{n-1}{n}\right) = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right) = 1 \rightarrow \left\{\frac{n-1}{n}\right\} \text{ is conv.}$

5. $\lim_{n \rightarrow \infty} 5 = 5 \rightarrow \{5\} \text{ is conv.}$

Theorem:

If $\lim_{n \rightarrow \infty} |a_n| = 0$ then $\lim_{n \rightarrow \infty} a_n = 0$

[Proof]:

we know: $-|a_n| \leq a_n \leq |a_n|$

now, since $\lim_{n \rightarrow \infty} |a_n| = 0$ (given)

$$\lim_{n \rightarrow \infty} -|a_n| = -\lim_{n \rightarrow \infty} |a_n| = 0$$

by Sandwich Thm. $\lim_{n \rightarrow \infty} a_n = 0$

$$6. \left\{ (-1)^{n+1} \frac{1}{n} \right\}_{n=1}^{\infty}$$

$$a_n = (-1)^{n+1} \frac{1}{n} \rightarrow |a_n| = \left| (-1)^{n+1} \frac{1}{n} \right| = \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} |a_n| = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\therefore \lim_{n \rightarrow \infty} a_n = 0 \rightarrow \{a_n\} \text{ conv. to } 0.$$

$$7. \lim_{n \rightarrow \infty} \frac{1}{n!} = 0 \rightarrow \left\{ \frac{1}{n!} \right\} \text{ is conv.}$$

$$8. \{(-1)^{n+1}\}$$

$$\lim_{n \rightarrow \infty} |a_n| = \lim_{n \rightarrow \infty} |(-1)^{n+1}| = 1 \neq 0$$

$$a_n = (-1)^{n+1} = \begin{cases} 1, & \text{if } n \text{ is odd} \\ -1, & \text{if } n \text{ is even} \end{cases}$$

$$\lim_{n \rightarrow \infty} a_n = DNE \quad (\text{because it has two different limit})$$

$$\therefore \{a_n\} \text{ is div.}$$

Theorem [Sandwich Theorem]:

Let $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$ be sequences of real number. If $\{a_n\} \leq \{b_n\} \leq \{c_n\}$ for all n beyond some index N , and if

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$$

$$\text{then, } \lim_{n \rightarrow \infty} b_n = L$$

Theorem:

Let $\{a_n\}$ and $\{b_n\}$ be sequences of real number and let A and B be real numbers. If $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$ then,

- $\lim_{n \rightarrow \infty} (a_n \pm b_n) = A \pm B$

- $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = AB$

- $\lim_{n \rightarrow \infty} (K \cdot b_n) = KB$, K is any real number

- $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{A}{B}$, $B \neq 0$

Some Basic Limits:-

$$1. \lim_{n \rightarrow \infty} \frac{\ln(n)}{n} = 0$$

$$2. \lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$$

$$3. \lim_{n \rightarrow \infty} (x)^{\frac{1}{n}} = 1, \quad x > 0$$

$$4. \lim_{n \rightarrow \infty} (x)^n = \begin{cases} 0, & -1 < x < 1 \\ 1, & x = 1 \\ \infty, & x > 1 \\ DNE, & x \leq -1 \end{cases}$$

$$5. \lim_{n \rightarrow \infty} \frac{(x)^n}{n!} = 0, \quad \text{for any } x$$

$$6. \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x, \quad \text{for any } x$$

$$\text{In general } \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^{yn} = e^{xy}, \quad x, y \in \mathbb{R}$$

$$7. \lim_{n \rightarrow \infty} \frac{\sin\left(\frac{k}{n}\right)}{\left(\frac{k}{n}\right)} = 1, \quad \text{where } k \in \mathbb{R} \text{ and } \left(\frac{k}{n}\right) \rightarrow 0$$

$$8. \lim_{n \rightarrow \infty} \frac{\tan\left(\frac{k}{n}\right)}{\left(\frac{k}{n}\right)} = 1$$

$$9. \lim_{n \rightarrow \infty} \frac{1 - \cos\left(\frac{k}{n}\right)}{\left(\frac{k}{n}\right)} = 0$$

Examples: find the limits of the following sequences and determine their converges?

1. $a_n = \frac{2n+1}{1-3\sqrt{n}}$

$$\begin{aligned}\lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \frac{2n+1}{1-3\sqrt{n}} = \lim_{n \rightarrow \infty} \frac{2\frac{n}{\sqrt{n}} + \frac{1}{\sqrt{n}}}{\frac{1}{\sqrt{n}} - 3} \\ &= \lim_{n \rightarrow \infty} \frac{2\sqrt{n} + \frac{1}{\sqrt{n}}}{\frac{1}{\sqrt{n}} - 3} = \frac{\infty + 0}{0 - 3} = -\infty \\ &\therefore \{a_n\} \text{ is div.}\end{aligned}$$

2. $a_n = \left(2 - \frac{1}{2^n}\right) \left(3 + \frac{1}{2^n}\right)$

$$\begin{aligned}\lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \left[\left(2 - \frac{1}{2^n}\right) \left(3 + \frac{1}{2^n}\right) \right] \\ &= \lim_{n \rightarrow \infty} \left(2 - \frac{1}{2^n}\right) \cdot \lim_{n \rightarrow \infty} \left(3 + \frac{1}{2^n}\right) \\ &= (2) \cdot (3) = 6 \\ &\therefore \{a_n\} \text{ is conv.}\end{aligned}$$

3. $\left\{ \frac{1}{\sqrt{n}} \tan^{-1} n \right\}$

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \tan^{-1} n &= \frac{\frac{\pi}{2}}{\infty} = 0 \\ &\therefore \{a_n\} \text{ is conv.}\end{aligned}$$

4. $\{\ln(n) - \ln(n + 1)\}$

$$\begin{aligned}\lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \ln\left(\frac{n}{n+1}\right) \\ &= \ln\left(\lim_{n \rightarrow \infty} \frac{1}{1 + \frac{1}{n}}\right) = \ln(1) = 0 \\ &\therefore \{a_n\} \text{ is conv.}\end{aligned}$$

5. $a_n = \frac{\ln(n)}{\ln(2n)}$

$$\begin{aligned}\lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \frac{\ln(n)}{\ln(2n)} = \lim_{n \rightarrow \infty} \frac{\ln(n)}{\ln(2) + \ln(n)} = \frac{\infty}{\infty} \\ &= \lim_{n \rightarrow \infty} \left(\frac{1}{\frac{\ln(2)}{\ln(n)} + 1} \right) = 1 \\ &\therefore \{a_n\} \text{ is conv.}\end{aligned}$$

6. $\{n - \sqrt{n^2 - 1}\}$

$$\begin{aligned}\lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} (n - \sqrt{n^2 - 1}) = \infty - \infty \\ &= \lim_{n \rightarrow \infty} (n - \sqrt{n^2 - 1}) \cdot \frac{n + \sqrt{n^2 - 1}}{n + \sqrt{n^2 - 1}} \\ &= \lim_{n \rightarrow \infty} \frac{n^2 - (n^2 - 1)}{n + \sqrt{n^2 - 1}} = \lim_{n \rightarrow \infty} \frac{1}{n + \sqrt{n^2 - 1}} = 0 \\ &\therefore \{a_n\} \text{ is conv.}\end{aligned}$$

$$7. a_n = n \left(1 - \cos \frac{1}{n} \right)$$

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} n \left(1 - \cos \frac{1}{n} \right) \\ &= \lim_{n \rightarrow \infty} \frac{\left(1 - \cos \frac{1}{n} \right)}{\left(\frac{1}{n} \right)} = 0 \\ &\therefore \{a_n\} \text{ is conv.} \end{aligned}$$

$$8. a_n = \frac{\sin^2 n}{2^n}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \frac{\sin^2 n}{2^n} \\ 0 &\leq \sin^2 n \leq 1 \\ \frac{0}{2^n} &\leq \frac{\sin^2 n}{2^n} \leq \frac{1}{2^n} \\ \lim_{n \rightarrow \infty} \frac{0}{2^n} &= \lim_{n \rightarrow \infty} \frac{1}{2^n} = 0 \\ \rightarrow \lim_{n \rightarrow \infty} \frac{\sin^2 n}{2^n} &= 0 \quad (\text{by S.T}) \\ &\therefore \{a_n\} \text{ is conv.} \end{aligned}$$

$$9. a_n = \frac{n!}{n^n}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \frac{n!}{n^n} = \frac{\infty}{\infty} \\ 0 &< \frac{n(n-1)(n-2)(n-3) \dots 3.2.1}{n.n.n.n \dots \dots \dots n.n.n} < \frac{1}{n} \\ \lim_{n \rightarrow \infty} 0 &= \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \quad \rightarrow \quad \lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0 \quad (\text{by S.T}) \\ &\therefore \{a_n\} \text{ is conv} \end{aligned}$$

Remark:

$$- \lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} a_{n-1} = L$$

- A sequence $\{a_n\}$ is called increasing if $a_n < a_{n+1}$ for all $n > 1$. It is called decreasing if $a_n > a_{n+1}$ for all $n > 1$.

- A sequence $\{a_n\}$ is bounded from above if there is a number M such that: $a_n \leq M$ for all n . The number M is called an upper bound of $\{a_n\}$.

- If M is an upper bound of $\{a_n\}$, but no number less than M is an upper bound of $\{a_n\}$, then M is the least upper bound for $\{a_n\}$.

- A sequence $\{a_n\}$ is bounded from below if there is a number m such that: $a_n \geq m$ for all n . The number m is called a lower bound of $\{a_n\}$.

- If m is a lower bound of $\{a_n\}$, but no number greater than m is a lower bound of $\{a_n\}$, then m is the greatest lower bound for $\{a_n\}$.

- If the $\{a_n\}$ bounded from above and below, then $\{a_n\}$ is called bounded otherwise unbounded.

Theorem:

- A non-decreasing seq. of real numbers conv. iff it is bounded from above.
- A non-increasing seq. of real numbers conv. iff it is bounded from below.

Def. [Monotonic]:

- The seq. $\{a_n\}$ is called monotonic if it is either increasing or decreasing.

Theorem [Monotonic seq.]:

- Every bounded + monotonic seq. is conv.

Examples

1. The seq. $2, 4, 6, \dots, 2n, \dots = \{2n\}$

The seq. is unbounded

2. The seq. $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots, \frac{n}{n+1}, \dots = \left\{ \frac{n}{n+1} \right\}$

The seq. is non-Decreasing (Inc.) and

$\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1 \rightarrow$ The seq. is bounded from above by 1

So the seq. $\left\{ \frac{n}{n+1} \right\}$ is conv.

3. *The seq. $5, 5, 5, 5, \dots, 5, \dots, \{5\}$*

The seq. is non-decreasing (or non-Inc.) and bounded from above (or below) by 5

So the seq. $\{5\}$ is conv.

4. *The seq. $\left\{\frac{4n+1}{n+1}\right\} = \frac{5}{2}, \frac{9}{3}, \frac{13}{4}, \dots$*

The seq. is non-decreasing and, $\lim_{n \rightarrow \infty} \frac{4n+1}{n+1} = 4$ then the seq. bounded from above by 4

So the seq. $\left\{\frac{4n+1}{n+1}\right\}$ conv.

5. *$\left\{\frac{3}{n+1}\right\}$ Is the seq. monotonic?*

The seq. is decreasing since

$$a_n = \frac{3}{n+1} > a_{n+1} = \frac{3}{(n+1)+1} = \frac{3}{n+2}, \quad \forall n \geq 1$$

So the seq. $\left\{\frac{3}{n+1}\right\}$ Is monotonic.

6. *$\left\{\frac{1}{n^2}\right\} = 1, \frac{1}{4}, \frac{1}{9}, \frac{1}{16}, \dots, \frac{1}{n^2}, \dots$*

The seq. is bounded from above by 1 and below by 0, so the seq. is bounded.

$$a_n = \frac{1}{n^2} > a_{n+1} = \frac{1}{(n+1)^2}, \quad \forall n \geq 1$$

This mean that the seq. is decreasing

$\Rightarrow$ The seq. is monotonic

$\Rightarrow$ The seq. is conv.

Exam Question:-

A conv. seq. is given by: $a_1 = \frac{1}{2}$, $a_{n+1} = \frac{1}{2} \left(a_n + \frac{1}{a_n} \right)$, $\forall n \geq 1$

a- Find the first 3 terms for $\{a_n\}_{n=1}^{\infty}$.

b- Find the limit of $\{a_n\}_{n=1}^{\infty}$.

Sol.

a- $a_1 = \frac{1}{2}$, $a_2 = \frac{1}{2} \left(a_1 + \frac{1}{a_1} \right) = \frac{1}{2} \left(\frac{5}{2} \right) = \frac{5}{4}$, $a_3 = \frac{1}{2} \left(\frac{5}{4} + \frac{4}{5} \right) = \frac{41}{40}$

b- Since $\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} a_n$

$$\frac{1}{2} \lim_{n \rightarrow \infty} \left(a_n + \frac{1}{a_n} \right) = \lim_{n \rightarrow \infty} a_n$$

$$\frac{1}{2} \left(L + \frac{1}{L} \right) = L$$

$$L + \frac{1}{L} = 2L$$

$$L^2 = 1$$

$$L = 1 \text{ or } -1 \rightarrow L = 1$$

($L = -1$) is rejected since the seq. is conv. so the limit must be unique and the seq. is increasing(and starting with positive terms $a_1 = \frac{1}{2}$), so it is impossible to reach negative number.

Exercises-11.1

$$1. a_n = \sqrt[n]{3n^3} \rightarrow (ans = 1)$$

$$2. a_n = (n+4)^{\frac{1}{n+4}} \rightarrow (ans = 1)$$

$$3. a_n = \left(\frac{1}{3}\right)^n + \frac{1}{\sqrt{2^n}} \rightarrow (ans = 0)$$

$$4. a_n = \frac{3^n 6^n}{2^{-n} n!} \rightarrow (ans = 0)$$

$$5. a_n = \frac{n!}{(10)^{6n}} \rightarrow (ans = \infty)$$

$$6. a_n = \left(\frac{3n+1}{3n-1}\right)^n \rightarrow (ans = e^{2/3})$$

$$7. a_n = \left(1 - \frac{1}{n^2}\right)^n \rightarrow (ans = 1)$$

$$8. a_n = \sqrt[n]{n^2 + n} \rightarrow (ans = 1)$$

$$9. a_n = \left(\frac{2n+1}{3n+2}\right)^n \rightarrow (ans = 0)$$

$$10. a_n = \left(\frac{-3}{5}\right)^n \rightarrow (ans = 0)$$

$$11. a_n = \left(\frac{e}{\pi}\right)^n \rightarrow (ans = 0)$$

$$12. a_n = \left(\frac{-3}{2}\right)^n \rightarrow (ans = DNE)$$

$$13. a_n = (-1)^n \rightarrow (ans = DNE)$$

$$14. \left\{ n^2 \sin\left(\frac{3}{n^2}\right) + \frac{\cos^2 n}{3^n} \right\} \rightarrow (ans = 3)$$

$$15. \left\{ \frac{n^2 \sin\left(\frac{2}{n}\right)}{\sin n} \right\} \rightarrow (ans = \infty)$$

$$16. \left\{ \frac{n^3 + 5n}{5n^3 + \cos n + n^{-2}} \right\} \rightarrow \left(ans = \frac{1}{5} \right)$$

$$17. \left\{ \frac{n + \sin n}{\sqrt{4n^2 + 1}} \right\} \rightarrow \left(ans = \frac{1}{2} \right)$$

$$18. \left\{ \frac{(2n-1)!}{(2n+3)!} \right\} \rightarrow (ans = 0)$$

$$19. \left\{ (n-2)! \sin \left(\frac{1}{n!} \right) \right\} \rightarrow (ans = 0)$$

$$20. \left\{ (2^n + 3^n)^{1/n} \right\} \rightarrow (ans = 3)$$

$$21. \left\{ \left(1 + \frac{3}{n} \right)^{2n} \right\} \rightarrow (ans = e^6)$$

$$22. \left\{ \left(1 - \frac{\ln 2}{n} \right)^n \right\} \rightarrow (ans = \frac{1}{2})$$

$$23. \left\{ \left(\frac{n}{n+3} \right)^{5n} \right\} \rightarrow (ans = e^{-15})$$

$$24. \left\{ \left(\frac{n+1}{n+3} \right)^n \right\} \rightarrow (ans = e^{-2})$$

$$25. \left\{ (-1)^n \frac{\ln(n)}{n} \right\} \rightarrow (ans = 0)$$

$$26. \left\{ (-1)^{n+1} \frac{2n^2+1}{n^2+5} \tan^{-1} n \right\} \rightarrow (ans = DNE)$$

$$27. \left\{ \frac{3^{n+1}+n^2}{(-3)^{n+2n}} \right\} \rightarrow (ans = DNE)$$

$$28. \left\{ \begin{array}{l} \frac{n^2}{2n+1} \sin \left(\frac{\pi}{n} \right) \text{ if } n \text{ is even} \\ n^2 \ln \left(1 + \frac{2}{n^2} \right) \text{ if } n \text{ is odd} \end{array} \right\} \rightarrow (ans = DNE)$$

$$29. \left\{ \frac{(-1)^{n+\sin n}}{n+1} \right\} \rightarrow (ans = 0)$$

$$30. \left\{ \sqrt[n]{2} + (-1)^n n^2 e^{-n} + n \tan \frac{\pi}{n} \right\} \rightarrow (ans = \pi)$$

11.2-Infinite Series

Def. [infinite series]:

An infinite series is the sum of an infinite seq. of numbers. It is an expression in the form:

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots = \sum_{n=1}^{\infty} a_n$$

[Convergence of a series]:

$$\text{let } s_1 = a_1$$

$$s_2 = a_1 + a_2$$

$$s_3 = a_1 + a_2 + a_3$$

$$s_n = a_1 + a_2 + a_3 + \cdots + a_n$$

$$- \lim_{n \rightarrow \infty} s_n = L \Rightarrow \{s_n\} \text{ conv.} \Rightarrow \sum_{n=1}^{\infty} a_n = L \text{ (conv.)}$$

$$- \lim_{n \rightarrow \infty} s_n = \begin{pmatrix} DNE \\ \pm \infty \end{pmatrix} \Rightarrow \{s_n\} \text{ div.} \Rightarrow \sum_{n=1}^{\infty} a_n \text{ no sum (div.)}$$

Note:

$$s_n = a_1 + a_2 + a_3 + \cdots + a_n$$

$$s_n = s_{n-1} + a_n$$

$$a_n = s_n - s_{n-1}$$

**Examples*:* discuss the convergence of the following series and find the sum (if the series conv.):

1. $\sum_{n=1}^{\infty} \tan^{-1}(n) - \tan^{-1}(n+1)$

Sol:

Type equation here.

$$s_n = (\tan^{-1} 1 - \tan^{-1} 2) + (\tan^{-1} 2 - \tan^{-1} 3) + (\tan^{-1} 3 - \tan^{-1} 4) + \dots + (\tan^{-1} n - \tan^{-1} (n+1))$$

$$s_n = \tan^{-1} 1 - \tan^{-1} (n+1)$$

$$\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} [\tan^{-1} 1 - \tan^{-1} (n+1)] = \frac{\pi}{4} - \frac{\pi}{2} = \frac{-\pi}{4}$$

$$\therefore \{s_n\} \text{ conv.} \Rightarrow \sum_{n=1}^{\infty} \tan^{-1}(n) - \tan^{-1}(n+1) \text{ is conv.}$$

$$\text{sum} = \frac{-\pi}{4}$$

2. $\sum_{n=1}^{\infty} \frac{4}{(4n-3)(4n+1)}$

Sol:

$$\frac{4}{(4n-3)(4n+1)} = \frac{A}{4n-3} + \frac{B}{4n+1}, \quad A=1, B=-1$$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{4n-3} - \frac{1}{4n+1}$$

$$= \left(1 - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{9}\right) + \left(\frac{1}{9} - \frac{1}{13}\right) + \dots + \left(\frac{1}{4n-3} - \frac{1}{4n+1}\right)$$

$$\therefore s_n = 1 - \frac{1}{4n+1}$$

$$\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} 1 - \frac{1}{4n+1} = 1 - 0 = 1$$

$$\therefore \{s_n\} \text{ conv.} \Rightarrow \sum_{n=1}^{\infty} \frac{4}{(4n-3)(4n+1)} \text{ conv.}$$

$sum = 1$

Geometric Series:

It is a series in the form:

$$a + ar + ar^2 + ar^3 + \dots + ar^{n-1} + \dots = \sum_{n=1}^{\infty} ar^{n-1}$$

- a is the first term, $a \neq 0$.

- r is the ratio.

- The G.S $\sum_{n=1}^{\infty} ar^{n-1}$ is conv. if $|r| < 1$, $\text{sum} = \frac{a}{1-r}$.

- The G.S $\sum_{n=1}^{\infty} ar^{n-1}$ is div. if $|r| \geq 1$, no sum.

Examples

1. $\sum_{n=0}^{\infty} (-1)^n \frac{5}{4^n}$

Sol:

$$\begin{aligned} &= \sum_{n=0}^{\infty} 5 \frac{(-1)^n}{4^n} = \sum_{n=0}^{\infty} 5 \left(\frac{-1}{4} \right)^n \\ a &= 5, \quad r = \frac{-1}{4} \rightarrow |r| = \frac{1}{4} < 1 \text{ (conv.)} \\ s &= \frac{a}{1-r} = \frac{5}{1 - \left(\frac{-1}{4} \right)} = 4 \end{aligned}$$

2. $\sum_{n=0}^{\infty} \frac{2^{n+3}}{3^n}$

Sol:

$$= \sum_{n=0}^{\infty} 2^3 \cdot \left(\frac{2}{3} \right)^n, \quad a = 8, r = \frac{2}{3} \rightarrow |r| < 1 \text{ (conv.)}$$

$$s = \frac{a}{1-r} = \frac{8}{1-\frac{2}{3}} = 24$$

$$3. \sum_{n=0}^{\infty} \frac{\cos n\pi}{5^n}$$

Sol:

$$\cos n\pi = (-1)^n$$

$$\sum_{n=0}^{\infty} \frac{\cos n\pi}{5^n} = \sum_{n=0}^{\infty} \frac{(-1)^n}{5^n} = \sum_{n=0}^{\infty} \left(\frac{-1}{5}\right)^n$$

$$a = 1, r = \frac{-1}{5} \rightarrow |r| < 1 \text{ (conv.)}$$

$$ssum = \frac{a}{1-r} = \frac{1}{1-\left(\frac{-1}{5}\right)} = \frac{5}{6}$$

$$4. \left(\frac{-2}{3}\right)^2 + \left(\frac{-2}{3}\right)^3 + \left(\frac{-2}{3}\right)^4 + \dots$$

Sol:

$$\left(\frac{-2}{3}\right)^2 + \left(\frac{-2}{3}\right)^3 + \left(\frac{-2}{3}\right)^4 + \dots = \sum_{n=2}^{\infty} \left(\frac{-2}{3}\right)^n$$

$$a = \frac{4}{9}, r = \frac{-2}{3} \rightarrow |r| < 1 \text{ (conv.)}$$

$$sum = \frac{\frac{4}{9}}{1 - \left(\frac{-2}{3}\right)} = \frac{4}{15}$$

5. Express the number $1.\overline{28}$ as a ratio of two integers.

Sol:

$$\begin{aligned}
1.\overline{28} &= 1.28\ 28\ 28\ 28\ 28\ \dots \\
&= 1 + \frac{28}{100} + \frac{28}{(100)^2} + \frac{28}{(100)^3} + \dots \\
&= 1 + \frac{28}{100} \left(1 + \frac{1}{100} + \frac{1}{(100)^2} + \frac{1}{(100)^3} + \dots \right) \\
&\left(1 + \frac{1}{100} + \frac{1}{(100)^2} + \frac{1}{(100)^3} + \dots \right) \rightarrow G.S \quad a = 1, r = \frac{1}{100} \\
sum &= \frac{1}{1 - \frac{1}{100}} = \frac{100}{99} \\
1.\overline{28} &= 1 + \frac{28}{100} \cdot \frac{100}{99} = \frac{127}{99}
\end{aligned}$$

6. Find the values of x for which the series

$\sum_{n=0}^{\infty} \left(\frac{-1}{2}\right)^n (x-3)^n$ Conv. and find the sum of the series.

Sol:

$$\sum_{n=0}^{\infty} \left(\frac{-1}{2}\right)^n (x-3)^n = \sum_{n=0}^{\infty} \left(\frac{-1}{2} (x-3)\right)^n$$

$$a = 1, r = \frac{-1}{2}(x-3)$$

In order to this series to be conv. $|r| < 1$

$$\Rightarrow \left| \frac{-1}{2}(x-3) \right| < 1$$
$$\Rightarrow |x-3| < 2$$

$$\Rightarrow 1 < x < 5, \quad x \in (1, 5)$$

$$S_s = \frac{a}{1-r} = \frac{1}{1+\frac{1}{2}(x-3)} = \frac{2}{x-1}$$

Def.:

- The series $\sum_{n=1}^{\infty} \frac{1}{n}$ is called harmonic series. It is div.
- The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is called P-series. It is conv. if $p > 1$ and div. if $p \leq 1$.

Examples

$$7. \sum_{n=1}^{\infty} \frac{-8}{n}$$

Sol:

$$\sum_{n=1}^{\infty} \frac{-8}{n} = \sum_{n=1}^{\infty} (-8) \frac{1}{n} = -8 \sum_{n=1}^{\infty} \frac{1}{n} \rightarrow \text{div. H.S}$$

$$8. \sum_{n=1}^{\infty} \frac{3}{\sqrt{n}}$$

Sol:

$$\sum_{n=1}^{\infty} \frac{3}{\sqrt{n}} = 3 \sum_{n=1}^{\infty} \frac{1}{n^{1/2}} \rightarrow \text{div. P.S}, \quad p = \frac{1}{2} < 1$$

9. $\sum_{n=1}^{\infty} \frac{e}{n\sqrt{n}}$

Sol:

$$\sum_{n=1}^{\infty} \frac{e}{n\sqrt{n}} = e \sum_{n=1}^{\infty} \frac{1}{n^{3/2}} \rightarrow \text{conv. P.S.}, \quad p = \frac{3}{2} > 1$$

Theorem:

If the series $\sum_{n=1}^{\infty} a_n$ conv. then $\lim_{n \rightarrow \infty} a_n = 0$ and the converse is not true.

Proof:

We know $a_n = s_n - s_{n-1}$, $\sum_{n=1}^{\infty} a_n$ conv. (given)

Then, $\lim_{n \rightarrow \infty} s_n = L$ exist by definition of series

$$\therefore \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (s_n - s_{n-1}) = L - L = 0$$

Test [1]: the n-th term test for diverges

If $\lim_{n \rightarrow \infty} a_n \neq 0$ then $\sum_{n=1}^{\infty} a_n$ div.

Examples

1. $\sum_{n=0}^{\infty} \frac{n!}{(1000)^n}$

Sol:

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n!}{(1000)^n} = \infty \neq 0$$

$\therefore$ by test[1] the series is div.

2. $\sum_{n=1}^{\infty} \frac{2^n + 4^n}{3^n + 4^n}$

Sol:

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{2^n + 4^n}{3^n + 4^n} = \lim_{n \rightarrow \infty} \frac{\left(\frac{2}{4}\right)^n + 1}{\left(\frac{3}{4}\right)^n + 1} = 1 \neq 0$$

$\therefore$ by test[1] the series is div.

$$3. \sum_{n=1}^{\infty} \frac{n+1}{\sqrt{4n^2+3}}$$

Sol:

$$\sqrt{4n^2+3} = \sqrt{4n^2 \left(1 + \frac{3}{4n^2}\right)} = 2n \sqrt{1 + \frac{3}{4n^2}}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n+1}{2n \sqrt{1 + \frac{3}{4n^2}}} = \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{n}}{2 \sqrt{1 + \frac{3}{4n^2}}} = \frac{1}{2} \neq 0$$

$\therefore$ by test[1] the series is div.

$$4. \sum_{n=1}^{\infty} \left(\frac{n}{n-1}\right)^n$$

Sol:

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left(\frac{n}{n-1}\right)^n = \lim_{n \rightarrow \infty} \left(\frac{1}{1 - \frac{1}{n}}\right)^n = e \neq 0$$

$\therefore$ by test[1] the series is div.

$$5. \sum_{n=1}^{\infty} \frac{e^n}{e^n + n}$$

$$\lim_{x \rightarrow \infty} a_n = \lim_{x \rightarrow \infty} \frac{e^x}{e^x + x} = \lim_{x \rightarrow \infty} \frac{e^x}{e^x + 1} \lim_{x \rightarrow \infty} \frac{e^x}{e^x} = 1 \neq 0$$

$\therefore$ by test[1] the series is div.

$$6. \sum_{n=1}^{\infty} \ln\left(\frac{n}{n+1}\right)$$

Sol:

$$\lim_{x \rightarrow \infty} a_n = \lim_{x \rightarrow \infty} \ln\left(\frac{n}{n+1}\right) = \ln\left(\lim_{x \rightarrow \infty} \frac{n}{n+1}\right) = \ln(1) = 0$$

$$a_n = \ln(n) - \ln(n+1)$$

$$s_n = (\ln 1 - \ln 2) + (\ln 2 - \ln 3) + \dots + (\ln(n) - \ln(n+1))$$

$$s_n = \ln 1 - \ln(n+1) = -\ln(n+1)$$

$$\lim_{n \rightarrow \infty} s_n = -\infty$$

$$\therefore \{s_n\} \text{ div.} \Rightarrow \sum_{n=1}^{\infty} \ln\left(\frac{n}{n+1}\right) \text{ div.}$$

Theorem:

If $\sum a_n = A$ and $\sum b_n = B$ are convergent series

- $\sum a_n \pm b_n = A \pm B$ is convergent.

- If k is a non-zero constant then $\sum ka_n = kA$ is convergent.

- Conv. $\pm$ conv. = conv.

- Conv. $\pm$ div. = div.

- Div. $\pm$ div. = may be conv. or div.

(e.g. $\sum_{n=1}^{\infty} (-1)^{2n} + \sum_{n=1}^{\infty} (-1)^{2n-1} = 0$ which is conv. series)

Example

$$\sum_{n=0}^{\infty} \left(\frac{5}{2^n} - \frac{1}{3^n} \right)$$

Sol:

$$= \sum_{n=0}^{\infty} 5 \left(\frac{1}{2} \right)^n - \sum_{n=0}^{\infty} \left(\frac{1}{3} \right)^n$$

This is two conv. G.S

And we know Conv. $\pm$ conv. = conv.

$$\therefore S = S_1 - S_2$$

$$= \frac{5}{1 - \frac{1}{2}} - \frac{1}{1 - \frac{1}{3}} = \frac{17}{2}$$

Exercises-11.2

$$1. \sum_{n=2}^{\infty} \frac{2}{4n^2-8n+3} \rightarrow (\text{ans. } s = 1)$$

$$2. \sum_{n=1}^{\infty} \frac{2}{4n^2-1} \rightarrow (\text{ans. } s = 1)$$

$$3. \sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{n}\right) \rightarrow (\text{ans. no sum})$$

$$4. \sum_{n=0}^{\infty} \tan^{-1}(n+2) - \tan^{-1}(n+1) \rightarrow \left(\text{ans. } s = \frac{\pi}{4}\right)$$

$$5. \sum_{n=2}^{\infty} \left(\frac{2}{n^2-1}\right) \rightarrow \left(\text{ans. } s = \frac{3}{2}\right)$$

$$6. \sum_{n=2}^{\infty} 2^{\frac{1}{n-1}} - 2^{\frac{1}{n+1}} \rightarrow (\text{ans. } s = \sqrt{2})$$

$$7. \sum_{n=1}^{\infty} (2)^{n-2} (3)^{1-n} \rightarrow \left(\text{ans. } s = \frac{3}{2}\right)$$

$$8. \sum_{n=1}^{\infty} \left(\frac{\pi}{\pi+1}\right)^n \rightarrow (\text{ans. } s = \pi)$$

$$9. \sum_{n=0}^{\infty} \left(\frac{e}{e+1}\right)^n \rightarrow (\text{ans. } s = e + 1)$$

$$10. \sum_{n=1}^{\infty} \frac{e^{n+1}}{2^n} \rightarrow (\text{ans. no sum})$$

$$11. \sum_{n=1}^{\infty} \frac{3^n - 2^n}{(-4)^n} \rightarrow \left(\text{ans. } s = \frac{-2}{21}\right)$$

$$12. \sum_{n=1}^{\infty} \frac{(x-3)^n}{2^{n+1}} \rightarrow \left(\text{ans. } s = \frac{\frac{1}{2}\left(\frac{x-3}{2}\right)}{1 - \left(\frac{x-3}{2}\right)}, x \in (1,5)\right)$$

13. $\sum_{n=1}^{\infty} \frac{2^n}{x^n} \rightarrow \left(\text{ans. } s = \frac{2/x}{1-2/x}, x \in (2, \infty) \cup (-\infty, -2) \right)$

14. $e^{-x} + e^{-2x} + e^{-3x} + \dots \rightarrow \left(\text{ans. } s = \frac{e^{-x}}{1-e^{-x}}, x \in (0, \infty) \right)$

15. If $\sum_{n=1}^{\infty} e^{nx} = 10$ find $x \rightarrow (\text{ans. } x = \ln 10 - \ln 11)$

16. Let $x = 0.\overline{794}$

a. Express x as a sum of G.S

b. Find integers m and n such that $x = \frac{m}{n}$

$\rightarrow \left(\text{ans. } x = \frac{794}{999} \right)$

11.3-Integral test

The Integral Test (Test [2])

Let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms, and let $f(x)$ be a function obtained by replacing n by x in the formula of a_n . If

- $f(x)$ is positive
- $f(x)$ is cont. on $[1, \infty)$
- $f(x)$ is decreasing for all $n \geq N$

(N : the smallest integer that satisfies the conditions.)

Then the series $\sum_{n=1}^{\infty} a_n$ and $\int_N^{\infty} f(x)dx$ are either both conv., or both div.

Note: when we use the integral test, it is not necessary to start the series or the integral with $n=1$.

Examples

1. $\sum_{n=4}^{\infty} \frac{\ln(n)}{\sqrt{n}}$

$$\text{let } m = \sqrt{n} \rightarrow m^2 = n$$
$$\therefore \sum_{n=4}^{\infty} \frac{\ln(n)}{\sqrt{n}} = \sum_{m=2}^{\infty} \frac{2 \ln(m)}{m}$$

$$\text{let } f(x) = \frac{2 \ln x}{x}$$

- $f(x)$ is positive (+ve)
- $f(x)$ is cont. on $[2, \infty)$
- $f(x)$ is decreasing because

$$f'(x) = \frac{2 \frac{1}{x} x - 2 \ln x}{x^2} = \frac{2(1 - \ln x)}{x^2} < 0, \forall x \geq 3$$

$$\therefore \int_3^{\infty} f(x) dx = \int_3^{\infty} \frac{2 \ln x}{x} dx = 2 \lim_{t \rightarrow \infty} \int_3^t \frac{\ln x}{x}$$

$$= \lim_{t \rightarrow \infty} [(\ln x)^2]_3^t = \infty$$

$$\int_3^{\infty} \frac{2 \ln x}{x} dx \text{ div.} \Rightarrow \sum_{n=4}^{\infty} \frac{\ln(n)}{\sqrt{n}} \text{ div.}$$

$$2. \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(\sqrt{n}+1)}$$

$$\text{let } f(x) = \frac{1}{\sqrt{x}(\sqrt{x}+1)} \quad \text{Positive and cont. } \forall x \geq 1$$

$$a_n = \frac{1}{\sqrt{n}(\sqrt{n}+1)} > a_{n+1} = \frac{1}{\sqrt{n+1}(\sqrt{n+1}+1)}$$

So $f(x)$ is decreasing $\forall x \geq 1$

$$\int_1^{\infty} \frac{1}{\sqrt{x}(\sqrt{x}+1)} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{\sqrt{x}(\sqrt{x}+1)} dx$$

$$= \lim_{t \rightarrow \infty} [2 \ln |\sqrt{n}+1|]_1^t = \infty$$

$$\int_1^{\infty} \frac{1}{\sqrt{x}(\sqrt{x}+1)} dx \text{ Div. } \Rightarrow \text{by Integral test } \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(\sqrt{n}+1)} \text{ div.}$$

$$3. \sum_{n=2}^{\infty} \frac{1}{n(\ln(n))^2}$$

$$\text{let } f(x) = \frac{1}{x(\ln x)^2} \quad , +ve \text{ and cont. } , \forall x \geq 2$$

$$a_n = \frac{1}{n(\ln(n))^2} > a_{n+1} = \frac{1}{(n+1)(\ln(n+1))^2} , \forall n \geq 2$$

So $f(x)$ is decreasing $\forall x \geq 2$

$$\int_2^{\infty} \frac{1}{x(\ln x)^2} dx = \lim_{t \rightarrow \infty} \int_2^t \frac{1}{x(\ln x)^2} dx$$

$$= \lim_{t \rightarrow \infty} \left[\frac{-1}{\ln x} \right]_2^t = \frac{1}{\ln 2}$$

$$\int_2^{\infty} \frac{1}{x(\ln x)^2} dx \text{ Conv. } \Rightarrow \text{by I.T } \sum_{n=2}^{\infty} \frac{1}{n(\ln(n))^2} \text{ conv.}$$

4. $\sum_{n=1}^{\infty} \text{sech}^2 n$

let $f(x) = \text{sech}^2 x$, +ve and cont., $\forall x \geq 1$

$$f'(x) = -2 \text{sech}^2 x \tanh x < 0, \forall x \geq 1$$

$\therefore f(x)$ decreasing $\forall x \geq 1$

$$\begin{aligned} \int_1^{\infty} \text{sech}^2 x \, dx &= \lim_{t \rightarrow \infty} \int_1^t \text{sech}^2 x \, dx = \lim_{t \rightarrow \infty} [\tanh x]_1^t \\ &= 1 - \left(\frac{e - e^{-1}}{e + e^{-1}} \right) \end{aligned}$$

$\int_1^{\infty} \text{sech}^2 x \, dx$ Conv. $\Rightarrow$ by I.T $\sum_{n=1}^{\infty} \text{sech}^2 n$ conv.

5. Find value of p for which $\sum_{n=2}^{\infty} \frac{1}{n (\ln(n))^p}$ converges?

let $f(x) = \frac{1}{x (\ln x)^p}$, +ve and cont. $\forall x \geq 2$

$$a_n = \frac{1}{n (\ln(n))^p} > a_{n+1} = \frac{1}{(n+1) (\ln(n+1))^p}, \forall n \geq 2$$

$\therefore f(x)$ decreasing $\forall x \geq 2$

$$\int_2^{\infty} \frac{1}{x (\ln x)^p} \, dx = \lim_{t \rightarrow \infty} \int_2^t \frac{1}{x (\ln x)^p} \, dx = \lim_{t \rightarrow \infty} \left[\frac{(\ln x)^{1-p}}{1-p} \right]_2^t$$

$$(\ln t)^{1-p} = 0 \text{ if } 1-p < 0 \rightarrow p > 1$$

$\therefore \sum_{n=2}^{\infty} \frac{1}{n (\ln(n))^p}$ Converges if $p > 1$

Exercises-11.3

1. $\sum_{n=2}^{\infty} \frac{1}{n \ln(n)} \rightarrow (ans = div.)$

2. $\sum_{n=2}^{\infty} \frac{1}{\ln(n)^n} \rightarrow (ans = div.)$

3. $\sum_{n=1}^{\infty} \frac{n}{e^{n^2}} \rightarrow (ans = conv.)$

4. $\sum_{n=1}^{\infty} \frac{1}{n^2} e^{\frac{1}{n}} \rightarrow (ans = conv.)$

5. $\sum_{n=1}^{\infty} \frac{e^{-\sqrt{n}}}{\sqrt{n}} \rightarrow (ans = conv.)$

6. $\sum_{n=2}^{\infty} \frac{\ln(n)}{n^2} \rightarrow (ans = conv.)$

7. $\sum_{n=1}^{\infty} \frac{1}{n(4+\ln^2(n))} \rightarrow (ans = conv.)$

8. $\sum_{n=2}^{\infty} \ln(\sqrt[n]{n}) \rightarrow (ans = div.)$

9. $\sum_{n=1}^{\infty} \frac{\ln(n)}{n} \rightarrow (ans = div.)$

10. Explain why integral test cannot be used to determine whether the series is convergent $\sum_{n=1}^{\infty} \frac{\cos(\pi n)}{\sqrt{n}}$

11. Find the values of p for which the series is convergent $\sum_{n=1}^{\infty} n(1+n^2)^p \rightarrow (ans. p < -1)$

11.4-The comparison test and limit comparison test

The comparison test (test [3])

Let $\sum a_n$, $\sum c_n$ and $\sum d_n$ be series with non-negative terms, suppose that for some integer N

$$d_n \leq a_n \leq c_n, \quad \forall n \geq N$$

- If $\sum c_n$ converges, then $\sum a_n$ also converges.
- If $\sum d_n$ diverges, then $\sum a_n$ also diverges.

Examples

1. $\sum_{n=1}^{\infty} \frac{\cos^2 n}{n^{3/2}}$

$$0 \leq \cos^2 n \leq 1$$
$$\frac{0}{n^{3/2}} \leq \frac{\cos^2 n}{n^{3/2}} \leq \frac{1}{n^{3/2}}$$

But $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ conv. p-series because $p = \frac{3}{2} > 1$

So by comparison test, the series $\sum_{n=1}^{\infty} \frac{\cos^2 n}{n^{3/2}}$ conv.

$$2. \sum_{n=1}^{\infty} \frac{2+\cos n}{\sqrt{n+8}}$$

$$-1 \leq \cos n \leq 1$$

$$1 \leq 2 + \cos n \leq 3$$

$$\frac{1}{\sqrt{n+8}} \leq \frac{2+\cos n}{\sqrt{n+8}} \leq \frac{3}{\sqrt{n+8}}$$

$$\left(\frac{1}{\sqrt{n+8n}} = \frac{1}{3\sqrt{n}} \right) \leq \frac{1}{\sqrt{n+8}} \leq \frac{2+\cos n}{\sqrt{n+8}} \leq \frac{3}{\sqrt{n+8}}$$

But $\frac{1}{3} \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ div. p-series $p=\frac{1}{2} < 1$, Then by C.T, the series $\sum_{n=1}^{\infty} \frac{2+\cos n}{\sqrt{n+8}}$ div.

$$3. \sum_{n=2}^{\infty} \frac{n+2}{n^2-n}$$

$$\left(\frac{n}{n^2} = \frac{1}{n} \right) < \frac{n}{n^2-n} < \frac{n+2}{n^2-n}$$

But $\sum_{n=1}^{\infty} \frac{1}{n}$ div. Harmonic series, Then by C.T the series

$$\sum_{n=2}^{\infty} \frac{n+2}{n^2-n} \text{ div.}$$

$$4. \sum_{n=1}^{\infty} \frac{10n+1}{n(n+1)(n+2)}$$

$$\frac{10n+1}{n(n+1)(n+2)} < \frac{10n+1}{n.n.n} < \left(\frac{10n+n}{n^3} = \frac{11}{n^2} \right)$$

But $\sum_{n=1}^{\infty} \frac{1}{n^2}$ conv. p-series $p=2 > 1$, Then by C.T the
series $\sum_{n=1}^{\infty} \frac{10n+1}{n(n+1)(n+2)}$ conv.

Remark:

- $n > \ln(n)$
- $\frac{1}{n} < \frac{1}{\ln(n)}$
- $\ln(n) > 1$, $\forall n \geq 3$

5. $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{2+\ln(n)}$

Sol:

We know: $n > \ln(n)$

$$\frac{2+n}{\sqrt{n}} > \frac{2+\ln(n)}{\sqrt{n}}$$
$$\left(\frac{\sqrt{n}}{2n+n} = \frac{\sqrt{n}}{3n} \right) < \frac{\sqrt{n}}{2+n} < \frac{\sqrt{n}}{2+\ln(n)}$$

But $\frac{1}{3} \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ is div. p-series $p=\frac{1}{2} < 1$, Then by C.T, the series $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{2+\ln(n)}$ div.

6. $\sum_{n=1}^{\infty} \frac{\ln(n+1)}{n+1}$

Sol:

We know: $\ln(n) > 1$, $\forall n \geq 3$

Then $\ln(n+1) > 1$, $\forall n \geq 2$

$$\left(\frac{1}{n+n} = \frac{1}{2n} \right) < \frac{1}{n+1} < \frac{\ln(n+1)}{n+1}$$

But $\frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n}$ is div. H.S, then by C.T $\sum_{n=1}^{\infty} \frac{\ln(n+1)}{n+1}$ div.

$$7. \sum_{n=1}^{\infty} \frac{\tanh n}{n^2}$$

Sol:

$$0 < \tanh n < 1, \quad \forall n + ve$$

$$\frac{0}{n^2} < \frac{\tanh n}{n^2} < \frac{1}{n^2}$$

But $\sum_{n=1}^{\infty} \frac{1}{n^2}$ conv. p-series $p=2 > 1$, Then by C.T the series

$$\sum_{n=1}^{\infty} \frac{\tanh n}{n^2} \text{ conv.}$$

$$8. \sum_{n=1}^{\infty} \frac{n+2^n}{n^2 \cdot 2^n}$$

Sol:

$$\frac{n+2^n}{n^2 \cdot 2^n} < \frac{2^n+2^n}{n^2 \cdot 2^n} < \left(\frac{2 \cdot 2^n}{n^2 \cdot 2^n} = \frac{2}{n^2} \right)$$

But $2 \sum_{n=1}^{\infty} \frac{1}{n^2}$ conv. p-series $p=2 > 1$, Then by C.T the

$$\text{series } \sum_{n=1}^{\infty} \frac{n+2^n}{n^2 \cdot 2^n} \text{ conv.}$$

$$9. \sum_{n=1}^{\infty} \frac{2^n + 3^n}{3^n + 4^n}$$

Sol:

$$\frac{2^n + 3^n}{3^n + 4^n} < \frac{2^n + 3^n}{4^n} < \left(\frac{3^n + 3^n}{4^n} = \frac{2 \cdot 3^n}{4^n} \right)$$

But $2 \sum_{n=1}^{\infty} \left(\frac{3}{4}\right)^n$ conv. G.S $|r| < 1$, Then by C.T the series

$$\sum_{n=1}^{\infty} \frac{2^n + 3^n}{3^n + 4^n} \text{ conv.}$$

10. $\sum_{n=1}^{\infty} \frac{n! \sqrt{n}}{(n+2)!}$

Sol:

$$\sum_{n=1}^{\infty} \frac{n! \sqrt{n}}{(n+2)!} = \sum_{n=1}^{\infty} \frac{n! \sqrt{n}}{(n+2)(n+1)n!} = \sum_{n=1}^{\infty} \frac{\sqrt{n}}{(n+2)(n+1)}$$

$$\frac{\sqrt{n}}{(n+2)(n+1)} < \left(\frac{\sqrt{n}}{n \cdot n} = \frac{1}{n^{3/2}} \right)$$

But $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ conv. p-series $p = \frac{3}{2} > 1$, Then by C.T the series $\sum_{n=1}^{\infty} \frac{n! \sqrt{n}}{(n+2)!}$ conv.

11. $\sum_{n=1}^{\infty} \frac{1}{n!}$

$$\left(\frac{1}{n!} = 1 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!} + \dots \right)$$

$$\left(\frac{1}{n!} = 1 + \frac{1}{2 \cdot 1} + \frac{1}{3 \cdot 2 \cdot 1} + \dots \right) < \left(\frac{1}{2^0} + \frac{1}{2^1} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}} \right)$$

$$\frac{1}{n!} < \frac{1}{2^{n-1}}, \quad \forall n$$

But $\sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = 2 \sum_{n=1}^{\infty} \left(\frac{1}{2} \right)^n$ conv. G.S $|r| = \frac{1}{2} < 1$, then by C.T the series $\sum_{n=1}^{\infty} \frac{1}{n!}$ is conv.

Limit comparison test (L.C.T)(test [4])

Suppose that $a_n > 0$ and $b_n > 0$, $\forall n \geq N$

- If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c > 0$, then $\sum a_n$ and $\sum b_n$ are both conv. or both div.

- If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ conv. , then $\sum a_n$ conv.

- If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ div. , then $\sum a_n$ div.

Examples

1. $\sum_{n=1}^{\infty} \frac{1}{n \sqrt[n]{n^3}}$

Sol:

Type equation here.

$$a_n = \frac{1}{n \sqrt[n]{n^3}}, \text{ and let } b_n = \frac{1}{n} \text{ (div. H.S)}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{n \sqrt[n]{n^3}}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{n}{n \sqrt[n]{n^3}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{n^3}} = 1 > 0$$

Then by L.C.T, the series $\sum_{n=1}^{\infty} \frac{1}{n^{\sqrt[n]{n^3}}}$ div.

2. $\sum_{n=1}^{\infty} \frac{\sqrt[n]{n}}{n^2}$

Sol:

$$a_n = \frac{\sqrt[n]{n}}{n^2}, \text{ and let } b_n = \frac{1}{n^2} \text{ (conv. p-series } p=2 > 1)$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\frac{\sqrt[n]{n}}{n^2}}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} \sqrt[n]{n} = 1 > 0$$

Then by L.C.T, the series $\sum_{n=1}^{\infty} \frac{\sqrt[n]{n}}{n^2}$ conv.

$$3. \sum_{n=1}^{\infty} \tan\left(\frac{1}{n}\right)$$

Sol:

$$a_n = \tan\left(\frac{1}{n}\right), \text{ and let } b_n = \frac{1}{n} \text{ (div. H.S)}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\tan\left(\frac{1}{n}\right)}{\left(\frac{1}{n}\right)} = 1 > 0$$

Then by L.C.T, the series $\sum_{n=1}^{\infty} \tan\left(\frac{1}{n}\right)$ div.

$$4. \sum_{n=2}^{\infty} \frac{1}{n\sqrt{n^2-1}}$$

Sol:

$$a_n = \frac{1}{n\sqrt{n^2-1}}, \text{ and let } b_n = \frac{1}{n^2} \text{ (conv. p-series } p=2 > 1)$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{a_n}{b_n} &= \lim_{n \rightarrow \infty} \frac{\frac{1}{n\sqrt{n^2-1}}}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} \frac{n}{\sqrt{n^2-1}} \\ &= \lim_{n \rightarrow \infty} \frac{n}{n\sqrt{1-\frac{1}{n^2}}} = 1 > 0 \end{aligned}$$

Then by L.C.T, the series $\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n^2-1}}$ conv.

$$5. \sum_{n=1}^{\infty} \frac{1}{1+\ln(n)}$$

Sol:

$$a_n = \frac{1}{1+\ln(n)}, \text{ and let } b_n = \frac{1}{n} \text{ (div. H.S)}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{1+\ln(n)}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{n}{1+\ln(n)} = \infty$$

Then by L.C.T, the series $\sum_{n=1}^{\infty} \frac{1}{1+\ln(n)}$ **div.**

$$6. \sum_{n=1}^{\infty} \frac{(\ln(n))^2}{n^3}$$

Sol:

$$a_n = \frac{(\ln(n))^2}{n^3}, \text{ and let } b_n = \frac{1}{n^2} \text{ (conv. p-series } p=2 > 1)$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{a_n}{b_n} &= \lim_{n \rightarrow \infty} \frac{\frac{(\ln(n))^2}{n^3}}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} \frac{(\ln(n))^2}{n} \\ &= \lim_{n \rightarrow \infty} \frac{2(\ln(n)) \cdot \frac{1}{n}}{1} = \lim_{n \rightarrow \infty} \frac{2(\ln(n))}{n} = \lim_{n \rightarrow \infty} \frac{\frac{2}{n}}{1} = 0 \end{aligned}$$

Then by L.C.T, the series $\sum_{n=1}^{\infty} \frac{(\ln(n))^2}{n^3}$ conv.

$$7. \sum_{n=1}^{\infty} 1 - \cos\left(\frac{1}{n}\right)$$

Sol:

$$a_n = 1 - \cos\left(\frac{1}{n}\right) \text{ And let } b_n = \frac{1}{n^2} \text{ (conv. p-series } p=2 > 1)$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{a_n}{b_n} &= \lim_{n \rightarrow \infty} \frac{1 - \cos\left(\frac{1}{n}\right)}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} \frac{\sin\left(\frac{1}{n}\right) \cdot \left(\frac{-1}{n^2}\right)}{\frac{-2}{n^3}} \\ &= \frac{1}{2} \lim_{n \rightarrow \infty} \frac{\sin\left(\frac{1}{n}\right)}{\left(\frac{1}{n}\right)} = \frac{1}{2} > 0 \end{aligned}$$

Then by L.C.T, the series $\sum_{n=1}^{\infty} 1 - \cos\left(\frac{1}{n}\right)$ conv.

Exercises-11.4

1. $\sum_{n=1}^{\infty} \frac{\cos^2 n}{3^n} \rightarrow (\text{ans.} = \text{conv.})$

2. $\sum_{n=1}^{\infty} \frac{\cos n}{n^2} \rightarrow (\text{ans.} = \text{conv.})$

3. $\sum_{n=1}^{\infty} \frac{2 - \cos n}{\sqrt{n-1}} \rightarrow (\text{ans.} = \text{div.})$

4. $\sum_{n=1}^{\infty} \frac{\sin \sqrt{n}}{\sqrt[3]{n^4+1}} \rightarrow (\text{ans.} = \text{conv.})$

5. $\sum_{n=3}^{\infty} \frac{1 + \sin^2 n}{\sqrt{n^2-4}} \rightarrow (\text{ans.} = \text{div.})$

6. $\sum_{n=2}^{\infty} \frac{2 + \cos n}{\sqrt[3]{n^2-1}} \rightarrow (\text{ans.} = \text{div.})$

7. $\sum_{n=1}^{\infty} \frac{\sin n + e^n}{3^n + n^2} \rightarrow (\text{ans.} = \text{conv.})$

8. $\sum_{n=2}^{\infty} \frac{\ln(n)}{n} \rightarrow (\text{ans.} = \text{div.})$

9. $\sum_{n=2}^{\infty} \frac{\ln(n)}{\sqrt{n}} \rightarrow (\text{ans.} = \text{div.})$

10. $\sum_{n=2}^{\infty} \frac{\ln(n)}{n^3 + n + 1} \rightarrow (\text{ans.} = \text{conv.})$

11. $\sum_{n=2}^{\infty} \frac{1}{\ln(n)} \rightarrow (\text{ans.} = \text{div.})$

12. $\sum_{n=2}^{\infty} \frac{(\ln(n))^3}{n} \rightarrow (\text{ans.} = \text{div.})$

13. $\sum_{n=1}^{\infty} \frac{1}{e^n + e^{-n}} \rightarrow (\text{ans.} = \text{conv.})$

14. $\sum_{n=1}^{\infty} \frac{1 + 2^n}{3^n + n^2} \rightarrow (\text{ans.} = \text{conv.})$

15. $\sum_{n=1}^{\infty} \frac{2n^2 + n + 1}{3n^3 + 3n + 7} \rightarrow (\text{ans.} = \text{div.})$

16. $\sum_{n=1}^{\infty} \frac{n + \sqrt{n}}{\sqrt[3]{n^7 + n^2}} \rightarrow (\text{ans.} = \text{conv.})$

17. $\sum_{n=1}^{\infty} \frac{3^{2n}}{(10)^n - 2n^{10}} \rightarrow (\text{ans.} = \text{conv.})$

$$18. \sum_{n=1}^{\infty} \frac{3n^2+5n}{e^n (n^2-1)} \rightarrow (ans. = conv.)$$

$$19. \sum_{n=1}^{\infty} \frac{n+3}{(n+1)\sqrt{n+2}} \rightarrow (ans. = div.)$$

$$20. \sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right) \rightarrow (ans. = div.)$$

$$21. \sum_{n=1}^{\infty} \cos\left(\frac{1}{n}\right) \rightarrow (ans. = div.)$$

$$22. \sum_{n=1}^{\infty} \sin\left(\frac{\pi}{n^2}\right) \rightarrow (ans. = conv.)$$

$$23. \sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{3^n}\right) \rightarrow (ans. = conv.)$$

$$24. \sum_{n=1}^{\infty} \frac{n^3+3^n}{n^2+4^n} \rightarrow (ans. = conv.)$$

$$25. \sum_{n=2}^{\infty} \ln\left(1 + \frac{1}{\sqrt{n}}\right) \rightarrow (ans. = div.)$$

$$26. \sum_{n=1}^{\infty} \tan\left(\frac{1}{n}\right) \rightarrow (ans. = div.)$$

$$27. \sum_{n=1}^{\infty} \frac{n \tan^{-1}(n)+1}{n^3-n+1} \rightarrow (ans. = conv.)$$

$$28. \sum_{n=1}^{\infty} \frac{n+\ln(n)}{n^2+1} \rightarrow (ans. = div.)$$

11.6-The ratio and n-th root test

The ratio test (test [5]):

Let $\sum a_n$ be a series with +ve terms ($a_n \geq 0$, $\forall n \geq N$)

And suppose that $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \rho$

, then

- The series $\sum a_n$ conv. if $\rho < 1$
- The series $\sum a_n$ div. if $\rho > 1$
- The test is inconclusive (fail) if $\rho = 1$

The n -th root test (test [6]):

Let $\sum a_n$ be a series with +ve terms ($a_n \geq 0$, $\forall n \geq N$)

And suppose that $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \rho$

, then

- The series $\sum a_n$ conv. if $\rho < 1$

- The series $\sum a_n$ div. if $\rho > 1$

- The test is inconclusive (fail) if $\rho = 1$

Examples

1. $\sum_{n=1}^{\infty} \frac{n!}{(10)^n}$

Sol:

$$a_n = \frac{n!}{(10)^n} , (a_n \geq 0 , \forall n \geq 1)$$

$$\therefore a_{n+1} = \frac{(n+1)!}{(10)^{n+1}} = \frac{(n+1)n!}{(10)^n \cdot 10}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+1)n!}{(10)^n \cdot 10} \cdot \frac{(10)^n}{n!} = \lim_{n \rightarrow \infty} \frac{(n+1)}{10} = \infty > 1$$

Therefore by Ratio test the series $\sum_{n=1}^{\infty} \frac{n!}{(10)^n}$ div.

2. $\sum_{n=2}^{\infty} \frac{n}{(\ln(n))^n}$

Sol:

$$a_n = \frac{n}{(\ln(n))^n} , (a_n \geq 0 , \forall n \geq 2)$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{n}{(\ln(n))^n}} = \lim_{n \rightarrow \infty} \frac{\sqrt[n]{n}}{\ln(n)} = \frac{1}{\infty} = 0 < 1$$

$$\therefore \sum_{n=2}^{\infty} \frac{n}{(\ln(n))^n} \text{ is conv. by n-th root test}$$

$$3 \cdot \sum_{n=1}^{\infty} (n!) \left(\frac{3}{n}\right)^n$$

Sol:

$$\begin{aligned} a_n &= n! \left(\frac{3}{n}\right)^n, \quad a_{n+1} = (n+1)! \left(\frac{3}{n+1}\right)^{n+1} \\ \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} &= \lim_{n \rightarrow \infty} \frac{(n+1)n! \cdot 3^n \cdot 3}{(n+1)^n \cdot (n+1)} \cdot \frac{n^n}{n! \cdot 3^n} = 3 \lim_{n \rightarrow \infty} \frac{n^n}{(n+1)^n} \\ &= 3 \lim_{n \rightarrow \infty} \left(\frac{1}{1+\frac{1}{n}}\right)^n = \frac{3}{e} > 1 \rightarrow \text{Div. by Ratio test} \end{aligned}$$

4. $\sum_{n=1}^{\infty} \frac{n!}{n^n}$

Sol:

$$a_n = \frac{n!}{n^n}, \quad a_{n+1} = \frac{(n+1)!}{(n+1)^{n+1}} = \frac{(n+1)n!}{(n+1)^n \cdot (n+1)}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+1)n!}{(n+1)^n \cdot (n+1)} \cdot \frac{n^n}{n!}$$

$$= \lim_{n \rightarrow \infty} \frac{n^n}{(n+1)^n} = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n = \lim_{n \rightarrow \infty} \left(\frac{1}{1 + \frac{1}{n}} \right)^n = \frac{1}{e} < 1$$

Then by Ratio test the series $\sum_{n=1}^{\infty} \frac{n!}{n^n}$ is conv.

5. $\sum_{n=1}^{\infty} \left(1 - \frac{1}{n}\right)^{n^2}$

Sol:

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \lim_{n \rightarrow \infty} \sqrt[n]{\left(1 - \frac{1}{n}\right)^{n^2}} = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n = \frac{1}{e} < 1$$

Then by n-th term test, the series $\sum_{n=1}^{\infty} \left(1 - \frac{1}{n}\right)^{n^2}$ conv.

6. $\sum_{n=1}^{\infty} \left(1 - \frac{1}{3n}\right)^n$

Sol:

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \lim_{n \rightarrow \infty} \sqrt[n]{\left(1 - \frac{1}{3n}\right)^n} = 1 \rightarrow \text{Fail}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{3n}\right)^n = e^{-\frac{1}{3}} \neq 0$$

Then by n-th term test (test [1]), the series is div.

1. apply
7

$$\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n) (3^n + 1)} \rightarrow a_n \quad +ve, \forall n \geq 1$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{\cancel{1 \cdot 3 \cdot 5 \cdots (2n-1)} \cdot \overset{(2n+1)}{(2n+2-1)}}{\cancel{2 \cdot 4 \cdot 6 \cdots (2n)} \cdot \underset{2(n+1)}{(2n+2)} \cdot \overset{(n+1)}{(3^{n+1} + 1)}} \cdot \frac{\cancel{2 \cdot 4 \cdot 6 \cdots (2n)} \cdot (2n) (3^n + 1)}{\cancel{1 \cdot 3 \cdot 5 \cdots (2n-1)}}$$

$$= \lim_{n \rightarrow \infty} \frac{2n+1}{2n+2} \cdot \lim_{n \rightarrow \infty} \frac{3^n + 1}{3^{n+1} + 1}$$

by L.R. $\leftarrow 1$

$$= \lim_{n \rightarrow \infty} \frac{3^n + 1}{3^n \cdot 3 + 1} = \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{3^n}}{3 + \frac{1}{3^n}} \xrightarrow{0} \frac{1}{3}$$

$$= \boxed{\frac{1}{3} < 1}$$

by ratio test, the series $\sum$ is Conv.

OR (another solution)

$$\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n) (3^n + 1)} < \frac{\cancel{2 \cdot 4 \cdot 6 \cdots 2n}}{\cancel{2 \cdot 4 \cdot 6 \cdots 2n} 3^n + 1} = \frac{1}{3^n + 1} < \frac{1}{3^n} = \left(\frac{1}{3}\right)^n$$

but $\sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n$ is Conv. ~~then~~ Geom. series
because $|r| = \frac{1}{3} < 1$

Then by C.T, the series is Conv.

Exercises-11.6

1. $\sum_{n=1}^{\infty} \frac{n! \ln(n)}{n (n+2)!} \rightarrow (\text{ans.} = \text{conv.})$

2. $\sum_{n=1}^{\infty} \frac{(n!)^n}{(n)^{n^2}} \rightarrow (\text{ans.} = \text{conv.})$

3. $\sum_{n=1}^{\infty} \frac{4^n (n!)^2}{(2n)!} \rightarrow (\text{ans.} = \text{div.})$

4. $a_1 = 3, a_{n+1} = \frac{n}{n+1} a_n \rightarrow (\text{ans.} = \text{div.})$

5. $\sum_{n=1}^{\infty} \frac{n^2}{3^n} \rightarrow (\text{ans.} = \text{conv.})$

6. $\sum_{n=1}^{\infty} \left(\frac{2}{n}\right)^n n! \rightarrow (\text{ans.} = \text{conv.})$

7. $\sum_{n=1}^{\infty} \frac{2^n n^{2n}}{(1+3n^2)^n} \rightarrow (\text{ans.} = \text{conv.})$

8. $\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^{n^2} \rightarrow (\text{ans.} = \text{conv.})$

9. $\sum_{n=1}^{\infty} 2^{-n} \left(1 + \frac{2}{n}\right)^{n^2} \rightarrow (\text{ans.} = \text{div.})$

10. $\sum_{n=1}^{\infty} \frac{(3n)!}{e^{2n}} \rightarrow (\text{ans.} = \text{div.})$

11. $\sum_{n=1}^{\infty} (1 + \cos^2 n) \frac{(n)^{2n}}{(2n+1)^n} \rightarrow (\text{ans.} = \text{div.})$

11.5-The Alternating series

Def.:

The alternating series is an expression of the form:

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n = a_1 - a_2 + a_3 - a_4 + \cdots + (-1)^{n+1} a_n + \cdots$$

Or

$$\sum_{n=1}^{\infty} (-1)^n a_n = -a_1 + a_2 - a_3 + a_4 + \cdots + (-1)^n a_n + \cdots$$

Theorem [convergence of the alternating series]

/ The alternating series test (test [7]):

The alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ conv. if the following three conditions are satisfied:

- The a_n 's are all positive
- The positive a_n 's are non-increasing, that is
 $a_n \geq a_{n+1}$, $\forall n \geq N$ for some integer N
- $\lim_{n \rightarrow \infty} a_n = 0$

Examples

1. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n}$

Sol:

- The a_n 's are all +ve, $\forall n \geq 1$

- $a_n = \frac{1}{n} \geq a_{n+1} = \frac{1}{n+1}$, $\forall n \geq 1$, Then a_n is decreasing
- $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$

Then by Alternating series test, the series is conv.

$$2. \sum_{n=1}^{\infty} (-1)^n \frac{2n+1}{3n+2}$$

Sol:

$$\lim_{n \rightarrow \infty} \frac{2n+1}{3n+2} = \frac{2}{3} \neq 0$$

$$\therefore \sum_{n=1}^{\infty} (-1)^n \frac{2n+1}{3n+2} \text{ is div. by test [1]}$$

Def. [Absolutely convergent]:

If the series of the absolute value $\sum |a_n|$ conv. then the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ is called absolutely convergent.

Def. [conditionally convergent]:

If the alternating series is not absolutely convergent but satisfies the three conditions of the theorem then the series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ is called conditionally convergent.

$$\sum (-1)^n a_n \rightarrow \sum |a_n| \rightarrow \begin{cases} \sum |a_n| \text{ div.} & \left\{ \begin{array}{l} \lim_{n \rightarrow \infty} a_n \neq 0 \rightarrow \text{div.} \\ 3 \text{ conditions satisfied} \rightarrow \text{C.C} \\ \sum |a_n| \text{ conv.} \rightarrow \text{Abs. conv.} \end{array} \right. \end{cases}$$

- Remark:
- ① $|(-1)^{n+1}| = |(-1)^n| = 1$
 - ② Abs. conv. means conv. and
Cond. conv. means conv.
 - ③ if the third condition of the previous theorem
is (not) satisfied that is (i.e., $\lim_{n \rightarrow \infty} a_n \neq 0$)
then the series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ is Div by Div.
test
 - ④ Cond. convergent means (not) Abs. conv.

Summary

$$\sum (-1)^n a_n$$

$$\sum |a_n| = \sum a_n$$

$$\sum |a_n| \text{ Div}$$

$$\lim_{n \rightarrow \infty} a_n \neq 0$$

$$\therefore \sum (-1)^n a_n$$

Div

by

divergent
test

$$\textcircled{1} \lim_{n \rightarrow \infty} a_n = 0$$

$$\textcircled{2} a_n \downarrow$$

$$\textcircled{3} a_n \text{'s } +ve$$

$$\therefore \sum (-1)^n a_n$$

Cond. conv.

(C.C)

$$\sum |a_n| \text{ Conv.}$$

$$\therefore \sum (-1)^n a_n$$

~~Cond~~

Abs. conv.

(A.C)

$$3. \sum_{n=1}^{\infty} (-1)^{n+1} \frac{4^n}{3^{n+1}}$$

Sol:

$$\sum_{n=1}^{\infty} \left| (-1)^{n+1} \frac{4^n}{3^{n+1}} \right| = \sum_{n=1}^{\infty} \frac{4^n}{3^{n+1}} = \frac{1}{3} \sum_{n=1}^{\infty} \left(\frac{4}{3} \right)^n$$

div. G.S because $|r| > 1$

$$\lim_{n \rightarrow \infty} \frac{4^n}{3^{n+1}} = \infty \neq 0$$

$\therefore \sum_{n=1}^{\infty} (-1)^{n+1} \frac{4^n}{3^{n+1}}$ is div.

$$4. \sum_{n=2}^{\infty} (-1)^{n+1} \frac{1}{n \ln(n)}$$

Sol:

$$\sum_{n=2}^{\infty} \left| (-1)^{n+1} \frac{1}{n \ln(n)} \right| = \sum_{n=2}^{\infty} \frac{1}{n \ln(n)}$$

Let $f(x) = \frac{1}{x \ln x}$ +ve and cont. $\forall x \geq 2$

$$a_n = \frac{1}{n \ln(n)} > a_{n+1} = \frac{1}{(n+1) \ln(n+1)} \quad \forall n \geq 2$$

So $f(x)$ is decreasing $\forall x \geq 2$

$$\begin{aligned} \int_2^{\infty} \frac{1}{x \ln x} dx &= \lim_{t \rightarrow \infty} \int_2^t \frac{1}{x \ln x} dx \\ &= \lim_{t \rightarrow \infty} [\ln(\ln x)]_2^t = \infty \end{aligned}$$

$$\int_2^{\infty} \frac{1}{x \ln x} dx \text{ Div.} \Rightarrow \text{by I.T } \sum_{n=2}^{\infty} \frac{1}{n \ln(n)} \text{ div.}$$

- The a_n 's are all +ve, $\forall n \geq 2$
- a_n is decreasing
- $\lim_{n \rightarrow \infty} \frac{1}{n \ln(n)} = 0$

Therefore $\sum_{n=2}^{\infty} (-1)^{n+1} \frac{1}{n \ln(n)}$ is C.C

$$5. \sum_{n=1}^{\infty} (-1)^n \frac{\sin n}{n^2}$$

Sol:

$$\sum_{n=1}^{\infty} \left| (-1)^{n+1} \frac{\sin n}{n^2} \right| = \sum_{n=1}^{\infty} \frac{|\sin n|}{n^2}$$

$$0 \leq |\sin n| \leq 1$$

$$0 \leq |\sin n| \leq \frac{1}{n^2}$$

But $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is conv. p-series $p=2 > 1$ by C.T

Thus $\sum_{n=1}^{\infty} (-1)^n \frac{\sin n}{n^2}$ is Abs. conv.

$$6. \sum_{n=1}^{\infty} \frac{(-2)^{n+1}}{n+5^n}$$

Sol:

$$\sum_{n=1}^{\infty} \left| \frac{(-2)^{n+1}}{n+5^n} \right| = \sum_{n=1}^{\infty} \frac{2^{n+1}}{n+5^n} = 2 \sum_{n=1}^{\infty} \frac{2^n}{n+5^n}$$

$$\frac{2^n}{n+5^n} < \frac{2^n}{5^n} = \left(\frac{2}{5}\right)^n$$

$$\sum_{n=1}^{\infty} \left(\frac{2}{5}\right)^n \text{ Is conv. G.S because } |r| < 1$$

$$\text{Thus } \sum_{n=1}^{\infty} \frac{(-2)^{n+1}}{n+5^n} \text{ is Abs. conv.}$$

$$7. \sum_{n=1}^{\infty} (-1)^n \frac{n^3}{2n^3+3}$$

Sol:

$$\sum_{n=1}^{\infty} \left| (-1)^n \frac{n^3}{2n^3+3} \right| = \sum_{n=1}^{\infty} \frac{n^3}{2n^3+3}$$

$$\lim_{n \rightarrow \infty} \frac{n^3}{2n^3+3} = \frac{1}{2} \neq 0$$

$$\text{Then by test [1], } \sum_{n=1}^{\infty} (-1)^n \frac{n^3}{2n^3+3} \text{ is div.}$$

$$8. \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2^n n!}{n^n}$$

Sol:

$$\sum_{n=1}^{\infty} \left| (-1)^{n+1} \frac{2^n n!}{n^n} \right| = \sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{2^{n+1} (n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{2^n n!} = 2 \lim_{n \rightarrow \infty} \frac{n^n}{(n+1)^n}$$

$$2 \lim_{n \rightarrow \infty} \left(\frac{1}{1 + \frac{1}{n}} \right)^n = \frac{2}{e} < 1, \sum_{n=1}^{\infty} \frac{2^n n!}{n^n} \text{ conv. by Ratio test}$$

Thus $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{2^n n!}{n^n}$ is Abs. conv.

9. $\sum_{n=1}^{\infty} (-1)^n \frac{\tan^{-1} n}{n\sqrt{n+1}}$

$$\sum_{n=1}^{\infty} \left| (-1)^n \frac{\tan^{-1} n}{n\sqrt{n+1}} \right| = \sum_{n=1}^{\infty} \frac{|\tan^{-1} n|}{n\sqrt{n+1}}$$

$$0 < |\tan^{-1} n| < \frac{\pi}{2}$$

$$0 < \frac{|\tan^{-1} n|}{n\sqrt{n+1}} < \frac{\frac{\pi}{2}}{n\sqrt{n+1}} = \frac{\pi}{2} \frac{1}{n^{3/2}}$$

But $\frac{\pi}{2} \sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ is conv. p-series $p = \frac{3}{2} > 1$ by C.T

Thus $\sum_{n=1}^{\infty} (-1)^n \frac{\tan^{-1} n}{n\sqrt{n+1}}$ is Abs. conv.

10. $\sum_{n=1}^{\infty} (-1)^n \frac{3^n (n!)^2}{(2n+1)!}$

ANS: CONV. BY R.T

SO, A.S is A.C

Estimating Sums

Theorem [Alternating series Estimation]:

If $s = \sum (-1)^{n+1} b_n$ is the sum of alternating series that satisfies:

- b_n 's are all +ve
- $b_n \geq b_{n+1}$, $\forall n \geq N$
- $\lim_{n \rightarrow \infty} b_n = 0$

Then $|R_n| = |s - s_n| \leq b_{n+1}$

Note: this theorem for conv. alternating series only!

Examples: show that the series is convergent. How many terms of the series do we need to add in order to find the sum of the indicated accuracy?

1. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} \quad (|\text{error}| < 0.01)$

$$b_n = \frac{1}{n^2}$$

- $b_n = \frac{1}{n^2}$ are all +ve , $\forall n \geq 1$

- $b_n = \frac{1}{n^2} \geq b_{n+1} = \frac{1}{(n+1)^2}$, $\forall n \geq 1$

- $\lim_{n \rightarrow \infty} \frac{1}{n^2} = 0$

Then $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ is conv. by A.S.T

Now,

$$b_{n+1} < 0.01$$

$$\frac{1}{(n+1)^2} < \frac{1}{100} \Leftrightarrow (n+1)^2 > 100, \forall n \geq 10$$

$$\text{So } \forall n \geq 10, |R_n| = |s - s_n| \leq 0.01$$

$$2. \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^4} \quad (|\text{error}| < 0.001)$$

$$b_n = \frac{1}{n^4}$$

- $b_n = \frac{1}{n^4}$ are all +ve , $\forall n \geq 1$
- $b_n = \frac{1}{n^4} \geq b_{n+1} = \frac{1}{(n+1)^4}$, $\forall n \geq 1$
- $\lim_{n \rightarrow \infty} \frac{1}{n^4} = 0$

Then $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^4}$ is conv. by A.S.T

Now,

$$b_{n+1} = \frac{1}{(n+1)^4} < \frac{1}{1000} \leftrightarrow (n+1)^4 > 1000, \forall n \geq 5$$

$$\text{So } \forall n \geq 5, |R_n| = |s - s_n| \leq 0.001$$

Exercises-11.5

1. $\sum_{n=1}^{\infty} (-1)^n \left(\sqrt{n + \sqrt{n}} - \sqrt{n} \right) \rightarrow (\text{ans.} = \text{div.})$

2. $\sum_{n=1}^{\infty} \frac{(-1)^n (2n)!}{2^n n! \cdot n} \rightarrow (\text{ans.} = \text{div.})$

3. $\sum_{n=1}^{\infty} (-1)^n \frac{1}{\sqrt[3]{n^2+2}} \rightarrow (\text{ans.} = \text{C. C})$

4. $\sum_{n=1}^{\infty} (-1)^n \frac{n}{n^2+1} \rightarrow (\text{ans.} = \text{C. C})$

5. $\sum_{n=1}^{\infty} \frac{\cos(n\pi)}{n^2+2} \rightarrow (\text{ans.} = \text{A. C})$

6. $\sum_{n=1}^{\infty} (-1)^n \frac{3n^2+5}{n^{7/2+n^2+7}} \rightarrow (\text{ans.} = \text{A. C})$

7. $\sum_{n=1}^{\infty} (-1)^n \frac{n! 2^n}{(2n)!} \rightarrow (\text{ans.} = \text{A. C})$

8. $\sum_{n=1}^{\infty} (-1)^n \left(\frac{n+3}{n+1} \right)^n \rightarrow (\text{ans.} = \text{div.})$

9. $\sum_{n=1}^{\infty} (-1)^n \frac{n+\ln(n)}{3^n} \rightarrow (\text{ans.} = \text{A. C})$

10. $\sum_{n=1}^{\infty} (-1)^n \frac{n^2}{5^n} \rightarrow (\text{ans.} = \text{A. C})$

11. $\sum_{n=1}^{\infty} \frac{\sin(n)}{4^n} \rightarrow (\text{ans.} = \text{A. C})$

12. $\sum_{n=1}^{\infty} \frac{2^n \cos(3n)}{\left(1+\frac{1}{n}\right)^{n^2}} \rightarrow (\text{ans.} = \text{A. C})$

Exercises-11.7

1. Determine whether the seq. $a_n = n + \frac{1}{n}$ is increasing, decreasing or not monotonic. Is this seq. bounded?
(Ans.: a_n is increasing, $\{a_n\}$ div. and unbounded)
2. Find $\lim_{n \rightarrow \infty} \sqrt[n]{2^n + 3^n}$ if it exists. (Ans.= 3)
3. Find all values of $\theta \in [0, \pi]$ for which the series $\sum_{n=1}^{\infty} \cos^n \theta$ is conv. and find its sum.
(Ans.: $\theta \in (0, \pi)$, $s = \frac{\cos \theta}{1 - \cos \theta}$)
4. Determine whether the series $\sum_{n=1}^{\infty} \frac{4}{n^2 + 2n}$ is conv. or div. by expressing its partial sum s_n as a telescoping sum. If it's conv. , find its sum (Ans.: conv. , sum= 3)
5. Test each series for conv.:
 - $\sum_{n=2}^{\infty} \frac{1}{n (\ln(n))^2}$ (Ans.: conv.)
 - $\sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{1}{\sqrt{n}}$ (Ans.: conv.)
6. Determine whether it is A.C , C.C , or div. :
 - $\sum_{n=1}^{\infty} \frac{\cos(n\pi)}{n^{3/4}}$ (Ans. : C.C)
 - $\sum_{n=1}^{\infty} \left(\frac{n+2}{n}\right)^{n^2}$ (Ans. : div.)

7. True or false :

- The seq. $1, \frac{1}{2}, 1, \frac{1}{4}, 1, \frac{1}{8}, 1, \frac{1}{16}, \dots$ conv. to 1 (Ans. : F)
- A conv. seq. $\{a_n\}$ that satisfies $a_{n+1} = \sqrt{1 + a_n}, n \geq 1$ has the limit $L = \frac{1+\sqrt{5}}{2}$ (Ans. : T)
- The series $\sum_{n=1}^{\infty} \frac{3^n}{p^n}$ conv. $\forall p, s.t : |p| \geq 3$ (Ans. : F)
- If $\sum_{n=0}^{\infty} a_n = A$ and $\sum_{n=0}^{\infty} b_n = B$ ($a_n, b_n > 0$) then $\sum_{n=0}^{\infty} a_n b_n = AB$ (Ans. : F)

8. Determine A.C, C.C, div.

- $\sum_{n=1}^{\infty} \frac{n^2 \sin(2n)}{1+3^n}$ (Ans. : A.C)
- $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$ (Ans. : C.C)

9. Determine whether the following series conv. or div. :

- $\sum_{n=1}^{\infty} \frac{\ln(n)}{n^{3/2}}$ (Ans. : conv.)
- $\sum_{n=1}^{\infty} n \tan\left(\frac{1}{\sqrt{n}}\right)$ (Ans. : div.)
- $\sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{n}{n^2+1}\right)$ (Ans. : conv.)
- $\sum_{n=2}^{\infty} \left(\frac{n+1}{3n-1}\right)^{3n}$ (Ans. : conv.)

10. Find power series representation for $f(x) = \frac{x^3}{(x-2)^2}$ and determine its radius of conv.

11.8-The power series

$$\sum_{n=0}^{\infty} a_n (x - c)^n \quad \text{Or} \quad \sum_{n=0}^{\infty} a_n x^n, x \neq 0$$

Note:

- we fixed x (treat it as constant)
- We want to find (radius / interval) of conv.
- We will use only two tests (Ratio / n-th Root)

Power series summery

I. If $x \in I = [a, b], (a, b), [a, b), (a, b]$

$$R = \frac{b - a}{2}$$
$$\lim_{n \rightarrow \infty} = 1$$

II. If $x = I = a$

$$R = 0$$
$$\lim_{n \rightarrow \infty} = \infty$$

III. If $x \in (-\infty, \infty)$

$$R = \infty$$
$$\lim_{n \rightarrow \infty} = 0$$

Examples

1. $\sum_{n=2}^{\infty} (-1)^n \frac{x^n}{4^n \ln(n)}$

Sol:

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \frac{|x^{n+1}| |x|}{4^{n+1} \ln(n+1)} \cdot \frac{4^n \ln(n)}{|x^n|} \\ &= \frac{1}{4} \lim_{n \rightarrow \infty} \frac{\ln(n)}{\ln(n+1)} \cdot |x| = \frac{|x|}{4} \end{aligned}$$

We know by Ratio test the series conv. if $\frac{|x|}{4} < 1$

$$-4 < x < 4$$

Put $x = -4$:

$$\sum_{n=2}^{\infty} (-1)^n \frac{(-4)^n}{4^n \ln(n)} = \sum_{n=2}^{\infty} \frac{(-1 \times -4)^n}{4^n \ln(n)} = \sum_{n=2}^{\infty} \frac{1}{\ln(n)}$$

$$\frac{1}{n} < \frac{1}{\ln(n)}, \sum_{n=2}^{\infty} \frac{1}{n} \text{ div. harmonic series}$$

$\therefore$ By C.T $\sum_{n=2}^{\infty} \frac{1}{\ln(n)}$ div.

Put $x = 4$:

$$\sum_{n=2}^{\infty} (-1)^n \frac{(4)^n}{4^n \ln(n)} = \sum_{n=2}^{\infty} \frac{(-1)^n}{\ln(n)}$$

$$\sum_{n=2}^{\infty} \left| \frac{(-1)^n}{\ln(n)} \right| = \sum_{n=2}^{\infty} \frac{1}{\ln(n)}$$

- The a_n 's are all +ve, $\forall n \geq 2$

- $a_n = \frac{1}{\ln(n)} \geq a_{n+1} = \frac{1}{\ln(n+1)}$, $\forall n \geq 2$, Then a_n is decreasing

$$- \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \rightarrow \sum_{n=2}^{\infty} \frac{1}{\ln(n)} \text{ conv. by A.S.T}$$

$$I = (-4, 4] , R = \frac{4 - (-4)}{2} = 4$$

$$2. \sum_{n=2}^{\infty} \frac{\ln(n)}{n 8^n} (x-1)^{3n}$$

Sol:

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \frac{\ln(n+1) |x-1|^{3n} |x-1|^3}{(n+1) 8^n \cdot 8} \cdot \frac{n 8^n}{\ln(n) \cdot |x-1|^{3n}} \\ &= \frac{1}{8} |x-1|^3 \lim_{n \rightarrow \infty} \frac{\ln(n)}{\ln(n+1)} \cdot \lim_{n \rightarrow \infty} \frac{n}{n+1} = \frac{1}{8} |x-1|^3 \end{aligned}$$

We know by Ratio test the series conv. if $\frac{1}{8} |x-1|^3 < 1$

$$-1 < x < 3$$

Put $x = -1$:

$$\sum_{n=2}^{\infty} \frac{\ln(n)}{n 8^n} (-2)^{3n} = \sum_{n=2}^{\infty} (-1)^n \frac{\ln(n)}{n}$$

$$\sum_{n=2}^{\infty} \left| (-1)^n \frac{\ln(n)}{n} \right| = \sum_{n=2}^{\infty} \frac{\ln(n)}{n} \rightarrow \text{Alternating series}$$

- The a_n 's are all +ve, $\forall n \geq 2$

- $a_n = \frac{\ln(n)}{n}$, Then a_n is decreasing Since $f'(x) < 0$

$$- \lim_{n \rightarrow \infty} \frac{\ln(n)}{n} = 0 \rightarrow \sum_{n=2}^{\infty} \frac{\ln(n)}{n} \text{ conv. by A.S.T}$$

Put $x = 3$:

$$\sum_{n=2}^{\infty} \frac{\ln(n)}{n 8^n} (2)^{3n} = \sum_{n=2}^{\infty} \frac{\ln(n)}{n}$$

$$\ln(n) > 1 , \forall n \geq 3$$

$$\frac{\ln(n)}{n} > \frac{1}{n} \rightarrow \text{But } \sum_{n=2}^{\infty} \frac{1}{n} \text{ is div. harmonic series}$$

$$\therefore \sum_{n=2}^{\infty} \frac{\ln(n)}{n} \text{ Is div. by C.T}$$

$$I = [-1, 3) \text{ , } R = \frac{3 - (-1)}{2} = 2$$

$$3. \sum_{n=1}^{\infty} \frac{n}{(2n)!} (x+1)^n$$

Sol:

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \frac{(n+1) |x+1|^n |x+1|}{(2n+2)(2n+1)(2n)!} \cdot \frac{(2n)!}{n |x+1|^n} \\ &= \frac{|x+1|}{2} \lim_{n \rightarrow \infty} \frac{1}{n(2n+1)} = 0 < 1 \rightarrow \text{conv.} \end{aligned}$$

$$I = (-\infty, \infty), R = \infty$$

$$4. \sum_{n=1}^{\infty} (-1)^n \frac{n^n}{n+1} (x-3)^n$$

Sol:

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \frac{(n+1)^n (n+1) |x-3|^n |x-3|}{n+2} \cdot \frac{n+1}{n^n |x-3|^n} \\ |x-3| \lim_{n \rightarrow \infty} \frac{(n+1)(n+1)}{n+2} \cdot \left(\frac{n+1}{n} \right)^n &= |x-3| \cdot \infty \cdot e = \infty \end{aligned}$$

Converges only at $x-3=0$

$$\rightarrow I = x = 3$$

$$R = 0$$

Exercises-11.8

$$1. \sum_{n=1}^{\infty} \frac{2^n}{n^2} (x+2)^n \rightarrow \left(\text{ans. } I = \left[-\frac{5}{2}, -\frac{3}{2} \right], R = \frac{1}{2} \right)$$

$$2. \sum_{n=2}^{\infty} \frac{x^n}{(\ln(n))^n} \rightarrow (\text{ans. } I = (-\infty, \infty), R = \infty)$$

$$3. \sum_{n=1}^{\infty} n \cdot 4^n (x-1)^{2n} \rightarrow \left(\text{ans. } I = \left(\frac{1}{2}, \frac{3}{2} \right), R = \frac{1}{2} \right)$$

$$4. \sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln(n) 3^{2n}} (2x+3)^n \rightarrow \left(\text{ans. } I = (-6, 3], R = \frac{9}{2} \right)$$

$$5. \sum_{n=1}^{\infty} \left(\frac{n}{2} \right)^n (x-3)^n \rightarrow (\text{ans. } R = 0, x = 3)$$

$$6. \sum_{n=1}^{\infty} \frac{n! x^n}{(2n)!} \rightarrow (\text{ans. } I = (-\infty, \infty), R = \infty)$$

$$7. \sum_{n=0}^{\infty} \frac{n x^n}{4^n (n^2+1)} \rightarrow (\text{ans. } I = [-4, 4), R = 4)$$

$$8. \sum_{n=1}^{\infty} \frac{(-1)^n (2x-3)^n}{2n-1} \rightarrow \left(\text{ans. } I = (1, 2], R = \frac{1}{2} \right)$$

$$9. \sum_{n=1}^{\infty} \frac{(x-3)^n}{5^n \sqrt{2n+1}} \rightarrow (\text{ans. } I = [-2, 8), R = 5)$$

11.9-Representation of functions as power series in power of “x”

$\frac{1}{1+u}$	$\sum_{n=0}^{\infty} (-1)^n u^n$	$ u < 1$ $I = (-1, 1)$	$R = 1$
$\frac{1}{1-u}$	$\sum_{n=0}^{\infty} u^n$	$ u < 1$ $I = (-1, 1)$	$R = 1$
$\ln(1+u)$	$\sum_{n=0}^{\infty} \frac{(-1)^n u^{n+1}}{n+1}$	$I = (-1, 1]$	$R = 1$
$\tan^{-1} u$	$\sum_{n=0}^{\infty} \frac{(-1)^n u^{2n+1}}{2n+1}$	$I = [-1, 1]$	$R = 1$
e^u	$\sum_{n=0}^{\infty} \frac{u^n}{n!}$	$I = \mathbb{R}$	$R = \infty$
$\sin u$	$\sum_{n=0}^{\infty} \frac{(-1)^n u^{2n+1}}{(2n+1)!}$	$I = \mathbb{R}$	$R = \infty$
$\sinh u$	$\sum_{n=0}^{\infty} \frac{u^{2n+1}}{(2n+1)!}$	$I = \mathbb{R}$	$R = \infty$
$\cos u$	$\sum_{n=0}^{\infty} \frac{(-1)^n u^{2n}}{(2n)!}$	$I = \mathbb{R}$	$R = \infty$
$\cosh u$	$\sum_{n=0}^{\infty} \frac{u^{2n}}{(2n)!}$	$I = \mathbb{R}$	$R = \infty$

Recall:

- $\sin x \cos x = \frac{1}{2} \sin(2x)$
- $\cos^2 x = \frac{1}{2} (1 + \cos(2x))$
- $\sin^2 x = \frac{1}{2} (1 - \cos(2x))$
- $\cosh^2 x = \frac{1}{2} (\cosh(2x) + 1)$
- $\sinh^2 x = \frac{1}{2} (\cosh(2x) - 1)$
- $\sinh x \cosh x = \frac{1}{2} \sinh(2x)$
- $a^u = e^{u \ln a}$

Examples

1. $f(x) = \frac{x}{4+x^2}$

Sol:

$$\begin{aligned} f(x) &= \frac{x}{4\left(1+\frac{x^2}{4}\right)} = \frac{x}{4} \cdot \frac{1}{1+\frac{x^2}{4}} \\ &= \frac{x}{4} \sum_{n=0}^{\infty} (-1)^n \left(\frac{x^2}{4}\right)^n, \quad \left|\frac{x^2}{4}\right| < 1 \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{4^{n+1}}, \quad |x| < 2 \\ I &= (-2, 2), \quad R = 2 \end{aligned}$$

$$2. f(x) = \int_0^x \frac{1}{2-t} dt$$

Sol:

$$\begin{aligned} \frac{1}{2-t} &= \frac{1}{2\left(1-\frac{t}{2}\right)} = \frac{1}{2} \cdot \frac{1}{1-\frac{t}{2}} \\ &= \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{t}{2}\right)^n, \quad \left|\frac{t}{2}\right| < 1 \\ &= \sum_{n=0}^{\infty} \frac{t^n}{2^{n+1}}, \quad |t| < 2 \\ \therefore f(x) &= \int_0^x \sum_{n=0}^{\infty} \frac{t^n}{2^{n+1}} dt = \sum_{n=0}^{\infty} \int_0^x \frac{t^n}{2^{n+1}} dt \\ &= \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} \int_0^x t^n dt = \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} \cdot \frac{x^{n+1}}{n+1} \end{aligned}$$

$$3. f(x) = \frac{x - \tan^{-1} x}{x^3}$$

Sol:

$$\begin{aligned} \tan^{-1} x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}, \quad |x| \leq 1 \\ \tan^{-1} x &= x + \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} \\ x - \tan^{-1} x &= -\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} \\ \frac{x - \tan^{-1} x}{x^3} &= -\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} \cdot \frac{1}{x^3} \\ \therefore \frac{x - \tan^{-1} x}{x^3} &= \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^{2n-2}}{2n+1} \end{aligned}$$

$$4. f(x) = \frac{1}{(2+3x)^2}$$

Sol:

$$\frac{1}{2+3x} = \frac{1}{2} \cdot \frac{1}{1+\frac{3x}{2}} = \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{3x}{2}\right)^n, \quad \left|\frac{3x}{2}\right| < 1$$

$$\therefore \frac{1}{2+3x} = \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{3}{2}\right)^n x^n, \quad |x| < \frac{2}{3}$$

Diff. both sides w.r.t "x"

$$\frac{-1}{(2+3x)^2} \cdot 3 = \frac{1}{2} \sum_{n=1}^{\infty} (-1)^n \left(\frac{3}{2}\right)^n n x^{n-1}$$

$$\therefore \frac{1}{(2+3x)^2} = \frac{-1}{6} \sum_{n=1}^{\infty} (-1)^n \left(\frac{3}{2}\right)^n n x^{n-1}$$

$$5. f(x) = \frac{1}{x^2+6x+13} \text{ find power series of } f(x) \text{ in power of } x+3$$

Sol:

$$f(x) = \frac{1}{x^2+6x+13} = \frac{1}{4+(x+3)^2} = \frac{1}{4} \cdot \frac{1}{1+\frac{(x+3)^2}{4}}$$

$$= \frac{1}{4} \sum_{n=0}^{\infty} (-1)^n \left(\frac{(x+3)^2}{4}\right)^n, \quad \left|\frac{(x+3)^2}{4}\right| < 1$$

$$\therefore f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n (x+3)^{2n}}{4^{n+1}}, \quad |x+3| < 2$$

$$I = (-5, -1), \quad R = 2$$

6. Find the sum of the following:

a- $\sum_{n=0}^{\infty} \frac{3^n}{2^n n!}$

Sol:

Type equation here.

$$\sum_{n=0}^{\infty} \frac{3^n}{2^n n!} = \sum_{n=0}^{\infty} \frac{\left(\frac{3}{2}\right)^n}{n!} = e^{\frac{3}{2}}$$

b- $\sum_{n=2}^{\infty} \frac{\left(\frac{3}{2}\right)^n}{n!}$

Sol:

$$\therefore \sum_{n=0}^{\infty} \frac{\left(\frac{3}{2}\right)^n}{n!} = e^{\frac{3}{2}}$$

$$\sum_{n=0}^{\infty} \frac{\left(\frac{3}{2}\right)^n}{n!} = 1 + \frac{3}{2} + \sum_{n=2}^{\infty} \frac{\left(\frac{3}{2}\right)^n}{n!}$$

$$\therefore \sum_{n=2}^{\infty} \frac{\left(\frac{3}{2}\right)^n}{n!} = e^{\frac{3}{2}} - 1 - \frac{3}{2} = e^{\frac{3}{2}} - \frac{5}{2}$$

Exercises-11.9

$$1. f(x) = \ln(3-x) \rightarrow \left(\text{ans.} = \ln 3 - \sum_{n=1}^{\infty} \frac{x^n}{3^n n} \right)$$

$$2. f(x) = x^2 \tan^{-1}(2x) \rightarrow \left(\text{ans.} = \sum_{n=0}^{\infty} (-1)^n \frac{2^{2n+1} x^{2n+3}}{2n+1} \right)$$

$$3. f(x) = \int_0^x e^{-\pi t^2} dt \rightarrow \left(\text{ans.} = \sum_{n=0}^{\infty} \frac{(-\pi)^n}{n!} \cdot \frac{x^{2n+1}}{2n+1} \right)$$

$$4. g(t) = \begin{cases} \frac{e^t-1}{t} & , \quad t \neq 0 \\ 1 & , \quad t = 0 \end{cases}, \quad f(x) = \int_0^x g(t) dt$$

$$\text{Find power series of } f(x) \rightarrow \left(\text{ans.} = \sum_{n=1}^{\infty} \frac{1}{n!} \cdot \frac{x^n}{n} \right)$$

$$5. f(x) = \frac{1}{(1-x)^3} \rightarrow \left(\text{ans.} = \frac{1}{2} \sum_{n=2}^{\infty} n(n-1)x^{n-2} \right)$$

$$6. f(x) = \frac{x^2}{(x-2)^2} \rightarrow \left(\text{ans.} = \sum_{n=1}^{\infty} \frac{n x^{n+1}}{2^{n+1}} \right)$$

$$7. f(x) = \frac{4}{3+2x-x^2} \text{ find power series of } f(x) \text{ in power of } x-1$$

$$\rightarrow \left(\text{ans.} = \sum_{n=0}^{\infty} \frac{(x-1)^{2n}}{4^n}, \quad (x-1)^2 < 4 \right)$$

$$8. f(x) = \frac{x}{4^x} \rightarrow \left(\text{ans.} = \sum_{n=0}^{\infty} \frac{(-1)^n (\ln 4)^n x^{n+1}}{n!} \right)$$

$$9. \text{Find sum } \sum_{n=1}^{\infty} (-1)^n \frac{1}{(n+1)2^{n+1}} \rightarrow \left(\text{ans.} = \ln\left(\frac{3}{2}\right) - \frac{1}{2} \right)$$

$$10. \text{ a- } f(x) = \ln(1+x)^{x^2} \rightarrow \left(\text{ans.} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+3}}{n+1} \right)$$

$$\text{ b - Use part a to find the sum } \sum_{n=1}^{\infty} \frac{n+2}{n \cdot 2^{n+1}} \rightarrow \left(\text{ans.} = \ln 2 + \frac{1}{2} \right)$$

11.10-Taylor and McLaurin series

- The **Taylor series** generated by the function f at the point a is:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a) (x-a)^n}{n!}$$
$$= f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a) (x-a)^2}{2!} + \dots + \frac{f^{(n)}(a) (x-a)^n}{n!} + \dots$$

- If $a=0$, we get the **McLaurin series** that is:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0) (x)^n}{n!}$$
$$= f(0) + \frac{f'(0) x}{1!} + \frac{f''(0) x^2}{2!} + \dots + \frac{f^{(n)}(0) x^n}{n!} + \dots$$

Remark:

- $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
- $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
- $1, -1, 1, -1, \dots, (-1)^{n+1}, \dots$
- $-1, 1, -1, 1, \dots, (-1)^n, \dots$
- $0, 2, 4, 6, 8, \dots, 2n, \dots$
- $1, 3, 5, 7, 9, \dots, 2n+1, \dots$

- “ n ” degree Taylor polynomial of $f(x)$ at $x = a$

$$T_n(x) = f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a) (x-a)^2}{2!} + \dots + \frac{f^{(n)}(a) (x-a)^n}{n!}$$

1. Find the Taylor series of $f(x) = 3^x$ at $a = 1$

Sol:

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(a) (x-a)^n}{n!} \\ &= f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a) (x-a)^2}{2!} + \dots + \frac{f^{(n)}(a) (x-a)^n}{n!} + \dots \\ &= f(1) + \frac{f'(1)(x-1)}{1!} + \frac{f''(1) (x-1)^2}{2!} + \dots + \frac{f^{(n)}(1) (x-1)^n}{n!} + \dots \\ &= 3 + 3\ln 3 (x-1) + \frac{3(\ln 3)^2}{2!} (x-1)^2 + \dots + \frac{3(\ln 3)^n}{n!} (x-1)^n + \dots \\ &= \sum_{n=0}^{\infty} \frac{3(\ln 3)^n}{n!} (x-1)^n \end{aligned}$$

Or

$$\begin{aligned} \text{Let } t &= x - a = x - 1 \\ x &= t + 1 \\ f(x) &= f(t + 1) = 3^{t+1} = 3 \cdot 3^t \\ e^t &= \sum_{n=0}^{\infty} \frac{t^n}{n!} \\ e^{t \ln 3} &= \sum_{n=0}^{\infty} \frac{(t \ln 3)^n}{n!} \\ 3 e^{t \ln 3} &= \sum_{n=0}^{\infty} 3 \frac{(t \ln 3)^n}{n!} \\ \therefore f(x) &= \sum_{n=0}^{\infty} 3(\ln 3)^n \frac{(x-1)^n}{n!} \end{aligned}$$

2. Find Taylor series of $f(x) = \sin x$ at $a = \frac{\pi}{2}$

Sol:

$$\text{Let } t = x - a = x - \frac{\pi}{2}$$

$$x = t + \frac{\pi}{2}$$

$$f(x) = f\left(t + \frac{\pi}{2}\right) = \sin\left(t + \frac{\pi}{2}\right)$$

$$\sin\left(t + \frac{\pi}{2}\right) = \sin t \cos \frac{\pi}{2} + \cos t \sin \frac{\pi}{2}$$

$$= \cos t$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{(2n)!}$$

$$\therefore f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n \left(t + \frac{\pi}{2}\right)^{2n}}{(2n)!}$$

3. Find the Taylor series of $f(x) = \ln x$ at $a = 2$

Sol:

$$\text{Let } t = x - a = x - 2$$

$$x = t + 2$$

$$f(x) = f(t + 2) = \ln(t + 2) = \ln(2 + t)$$

$$= \ln\left(2\left(1 + \frac{t}{2}\right)\right) = \ln(2) + \ln\left(1 + \frac{t}{2}\right)$$

$$= \ln(2) + \sum_{n=0}^{\infty} \frac{(-1)^n \left(\frac{t}{2}\right)^{n+1}}{n+1}$$

$$\therefore f(x) = \ln(2) + \sum_{n=0}^{\infty} \frac{(-1)^n (x-2)^{n+1}}{2^{n+1} (n+1)}$$

4. Let $f(x) = \sqrt{1+x}$ find Taylor polynomial of degree 3 of $f(x)$ at $a = 0$

Or find the first 4 terms of McLaurin series of $f(x) = \sqrt{1+x}$

Sol:

$$\begin{aligned} T_3(x) &= f(0) + \frac{f'(0)x}{1!} + \frac{f''(0)x^2}{2!} + \frac{f^{(3)}(0)x^3}{3!} \\ &= 1 + \frac{1}{2}x - \frac{1}{4}\frac{x^2}{2!} + \frac{3}{8}\frac{x^3}{3!} \end{aligned}$$

5. Find McLaurin series of $f(x) = \frac{x^5}{(1+x^4)^2}$

Sol:

$$\frac{1}{1+x^4} = \sum_{n=0}^{\infty} (-1)^n (x^4)^n = \sum_{n=0}^{\infty} (-1)^n x^{4n}, \quad |x| < 1$$

Diff. both sides w.r.t "x"

$$\frac{-1}{(1+x^4)^2} \cdot 4x^3 = \sum_{n=1}^{\infty} (-1)^n 4n x^{4n-1}$$

$$\frac{x^3}{(1+x^4)^2} = \sum_{n=1}^{\infty} (-1)^{n+1} n x^{4n-1}$$

Multiply both sides by " x^2 "

$$f(x) = \frac{x^5}{(1+x^4)^2} = \sum_{n=1}^{\infty} (-1)^{n+1} n x^{4n+1}$$

The binomial series:

If k is any real number and $|x| < 1$, then

$$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n$$
$$= 1 + kx + \frac{k(k-1)}{2!} x^2 + \frac{k(k-1)(k-2)}{2!} x^3 + \dots$$

$$\text{Where } \binom{k}{n} = \frac{k(k-1)(k-2)\dots(k-n+1)}{n!}$$

6. Find the McLaurin series of the function

$$f(x) = \frac{1}{\sqrt{4-x}}, \text{ and its radius of convergence}$$

sol:

Type equation here.

$$\begin{aligned} f(x) &= \frac{1}{\sqrt{4-x}} = \frac{1}{\sqrt{4\left(1-\frac{x}{4}\right)}} = \frac{1}{\sqrt{4}} \frac{1}{\sqrt{1-\frac{x}{4}}} \\ &= \frac{1}{2} \left(1 + \left(-\frac{x}{4}\right)\right)^{-\frac{1}{2}} \\ &= \frac{1}{2} \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} \left(-\frac{x}{4}\right)^n, \quad \left|-\frac{x}{4}\right| < 1 \\ &\quad |x| < 4 \end{aligned}$$

$$-4 < x < 4$$

$$R = \frac{4 - (-4)}{2} = 4$$

7. - Find McLaurin series of $f(x) = x^3 e^{\frac{x}{2}}$

sol:

$$\text{We know } e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$e^{\frac{x}{2}} = \sum_{n=0}^{\infty} \frac{\left(\frac{x}{2}\right)^n}{n!} = \sum_{n=0}^{\infty} \frac{x^n}{2^n n!}$$

$$\therefore f(x) = x^3 e^{\frac{x}{2}} = \sum_{n=0}^{\infty} \frac{x^{n+3}}{2^n n!}$$

- Find sum: $\sum_{n=0}^{\infty} \frac{n+3}{2^n n!}$

Sol:

$$f'(x) = \sum_{n=0}^{\infty} \frac{(n+3) x^{n+2}}{2^n n!}$$

Put $x = 1$:

$$f'(1) = \sum_{n=0}^{\infty} \frac{(n+3)}{2^n n!}$$

$$\text{Also } f'(x) = 3x^2 e^{\frac{x}{2}} + x^3 e^{\frac{x}{2}} \frac{1}{2}$$

$$f'(1) = 3 e^{\left(\frac{1}{2}\right)} + \frac{1}{2} e^{\frac{1}{2}} = \frac{7}{2} e^{\frac{1}{2}}$$

$$\therefore \sum_{n=0}^{\infty} \frac{(n+3)}{2^n n!} = \frac{7}{2} e^{\frac{1}{2}}$$

Exercises-11.10

1. Find the McLaurin series of the function

$f(x) = \int_0^x \sin(t^2) dt$, and state its radius of convergence

$$\left(\text{ans.} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{4n+3}}{(2n+1)!(4n+3)} , R = \infty \right)$$

2. Find the sum of the series $\sum_{n=1}^{\infty} (-1)^n \frac{\pi^{2n+1}}{4^{2n+1} (2n+1)!}$ (Hint: use the McLaurin series for $\sin x$) $\rightarrow \left(\text{ans.} = \frac{\sqrt{2}}{2} - \frac{\pi}{4} \right)$

3. Find the Taylor poly. Of degree 3 of the $f(x) = \sqrt{1+x}$ centered at $a=0 \rightarrow \left(\text{ans.} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} \right)$

4. Find the forth McLaurin poly. For $f(x) = \sin(2x)$
 $\left(\text{ans.} = 2x - \frac{4}{3}x^3 \right)$

5. Find the Taylor series for $f(x) = \ln(x)$ at $a=2$
 $\left(\text{ans.} = \ln(2) + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(x-2)^n}{2^n n} \right)$

6. Find the McLaurin series for $f(x) = e^{5x}$ at $a=0$
 $\left(\text{ans.} = \sum_{n=0}^{\infty} 5^n \frac{x^n}{n!} \right)$

7. Find the Taylor series for $f(x) = x^4 + x - 2$ at $a=1$

$$(ans. = 5(x-1) + 6(x-1)^2 + 4(x-1)^3 + (x-1)^4)$$

8. Find the first 4 terms in the Taylor series for

$$(x-1)e^x \text{ near } x=1$$

$$(ans. = (x-1)e + (x-1)^2e + \frac{(x-1)^3}{2}e + \frac{(x-1)^4}{6}e)$$

9. Find the first 3 terms in the McLaurin series for $\sin^2 x$

$$(ans. = x^2 - \frac{x^4}{3} + \frac{2x^6}{45})$$

10. Find the Taylor series of $f(x) = \ln(1+x)$ at $x=0$

$$(ans. = \sum_{n=0}^{\infty} (-1)^{n+1} \frac{x^{n+1}}{n+1})$$

11. Find the Taylor poly. Of degree 3 for $f(x) = \sec x$ at $x = \frac{\pi}{4}$

$$(ans. T_3 = \sqrt{2} + \frac{\sqrt{2}}{1!} \left(x - \frac{\pi}{4}\right) + \frac{3\sqrt{2}}{2!} \left(x - \frac{\pi}{4}\right)^2 + \frac{11\sqrt{2}}{3!} \left(x - \frac{\pi}{4}\right)^3)$$

12. Find the Taylor series of $f(x) = \frac{2}{3^{x-4}}$ about $x=2$ and find its radius of convergence

$$(ans. = \sum_{n=0}^{\infty} (-1)^{n+1} \frac{3^n (x-2)^n}{2^n}, \quad R = \frac{2}{3})$$